

CSE 4415:: Wireless Networks

MAC Efficiency Boost:
Achieving Collision-Free Channel Access
Using Off-the-Shelf Technologies

Lecture 7.2

Solution Approach -1

(Discussed in Lecture-7.1)

Improvements of MAC Layer Inefficiency

:: Frequency Domain Backoff ::

A distributed approach that constructs a collision-free TDMA-like schedule using frequency-domain backoff, minimizing channel wastage, with cost of Additional Antenna and specific PHY layer techniques (OFDM).

Solution Approach-2

A Distributed approach for preparing Collision-free TDMA Schedule resulting Minimum Channel Wastage with cost of Multiple Round Learning phases

Paper Title: Winning the Lottery: Learning Perfect Coordination with Minimal Feedback

Publisher: *EEE Journal of Selected Topics in Signal Processing* (2013) GLOBECOM 2013
Wireless Communications Symposium

- Perfect Coordination (PC) protocol rely heavily on learning, exploiting the possibility to use both actions and silence as messages and the ability of stations to learn from their own histories while simultaneously enabling the learning of other stations.

Perfect Coordination (PC) protocol : Learning Phase (while Number of Nodes are known to All)

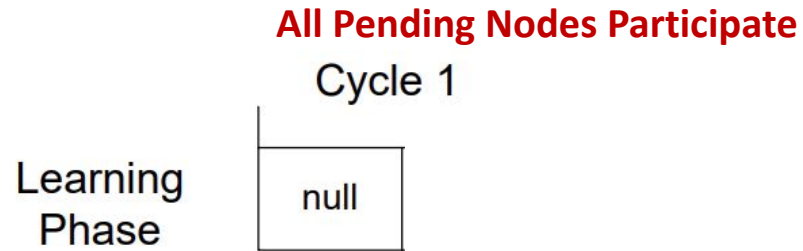


Fig. 1. Illustration of PC protocol when N is known. ($N = 3, K = 4$)

Perfect Coordination (PC) protocol : Learning Phase (while Number of Nodes are known to All)

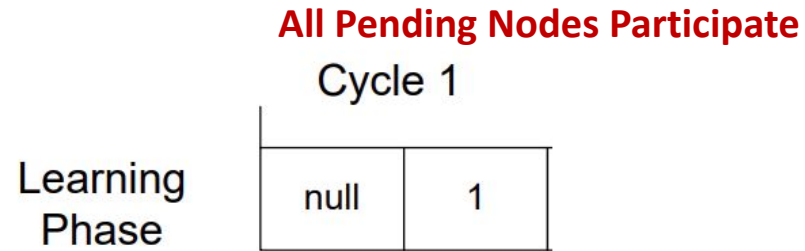


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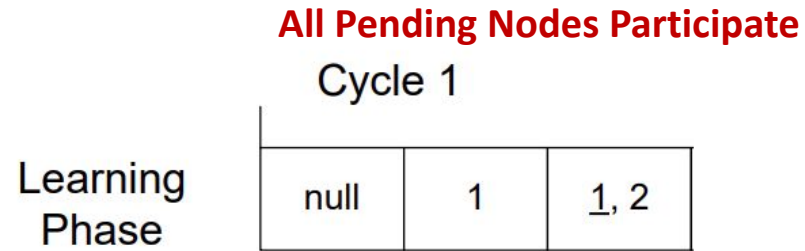


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Perfect Coordination (PC) protocol : Learning Phase (while Number of Nodes are known to All)

All Pending Nodes Participate

Cycle 1

Learning Phase				
	null	1	<u>1</u> , 2	<u>1</u> , 3

Fig. 1. Illustration of PC protocol when N is known. ($N = 3, K = 4$)

Perfect Coordination (PC) protocol : **Learning Phase (while Number of Nodes are known to All)**

Learning Phase	All Pending Nodes Participate Cycle 1				All Pending Nodes Participate Cycle 2	
	null	1	<u>1</u> , 2	<u>1</u> , 3	2, 3	

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	null	1	<u>1</u> , 2	<u>1</u> , 3	2, 3	null

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	null	1	<u>1</u> , 2	<u>1</u> , 3	2, 3	null	3

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Learning Phase	All Pending Nodes Participate Cycle 1				All Pending Nodes Participate Cycle 2			
	null	1	<u>1</u> , 2	<u>1</u> , 3	2, 3	null	3	<u>3</u> , 2

Fig. 1. Illustration of PC protocol when N is known. ($N = 3, K = 4$)

Perfect Coordination (PC) protocol : Learning Phase (while Number of Nodes are known to All)

	All Pending Nodes Participate				All Pending Nodes Participate				All Pending Nodes Participate			
	Cycle 1				Cycle 2				Cycle 3			
Learning Phase	null	1	<u>1</u> , 2	<u>1</u> , 3	2, 3	null	3	<u>3</u> , 2	null			

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Perfect Coordination (PC) protocol : Learning Phase (while Number of Nodes are known to All)

	All Pending Nodes Participate				All Pending Nodes Participate				All Pending Nodes Participate			
	Cycle 1				Cycle 2				Cycle 3			
Learning Phase	null	1	<u>1</u> , 2	<u>1</u> , 3	2, 3	null	3	<u>3</u> , 2	null	2		

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Perfect Coordination (PC) protocol : Learning Phase (while Number of Nodes are known to All)

	All Pending Nodes Participate				All Pending Nodes Participate				All Pending Nodes Participate			
	Cycle 1				Cycle 2				Cycle 3			
Learning Phase	null	1	<u>1</u> , 2	<u>1</u> , 3	2, 3	null	3	<u>3</u> , 2	null	2	<u>2</u>	<u>2</u>

Fig. 1. Illustration of PC protocol when N is known. ($N = 3, K = 4$)

Perfect Coordination (PC) protocol : Learning Phase (while Number of Nodes are known to All)

	Cycle 1				Cycle 2				Cycle 3			
Learning Phase	null	1	<u>1</u> , 2	<u>1</u> , 3	2, 3	null	3	<u>3</u> , 2	null	2	<u>2</u>	<u>2</u>
Transmisson Phase	1	3	2	1	3	2	...					

Fig. 1. Illustration of PC protocol when N is known. ($N = 3, K = 4$)

PC: Learning Phase
(while Number of Nodes are known to All)

PC: Learning Phase (while Number of Nodes are known to All)

All Pending Nodes Participate

Learning-to-Win				
Cycle 1	null	1	1, 2	1, 3

Learning
Phase

Fig. 2. Illustration of PC protocol when N is known. ($N = 3, K = 4, N^{max} = 4$)

PC: Learning Phase (while Number of Nodes are **known to All**)

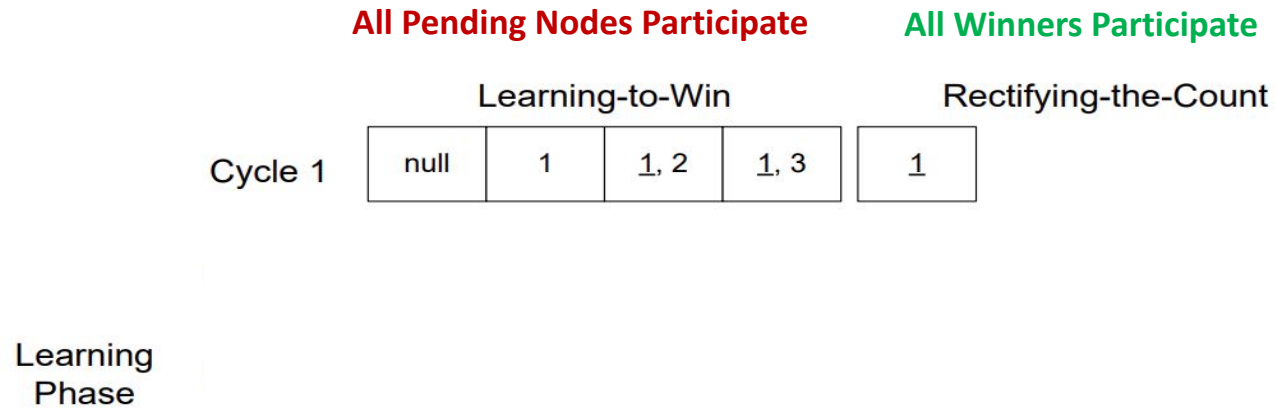


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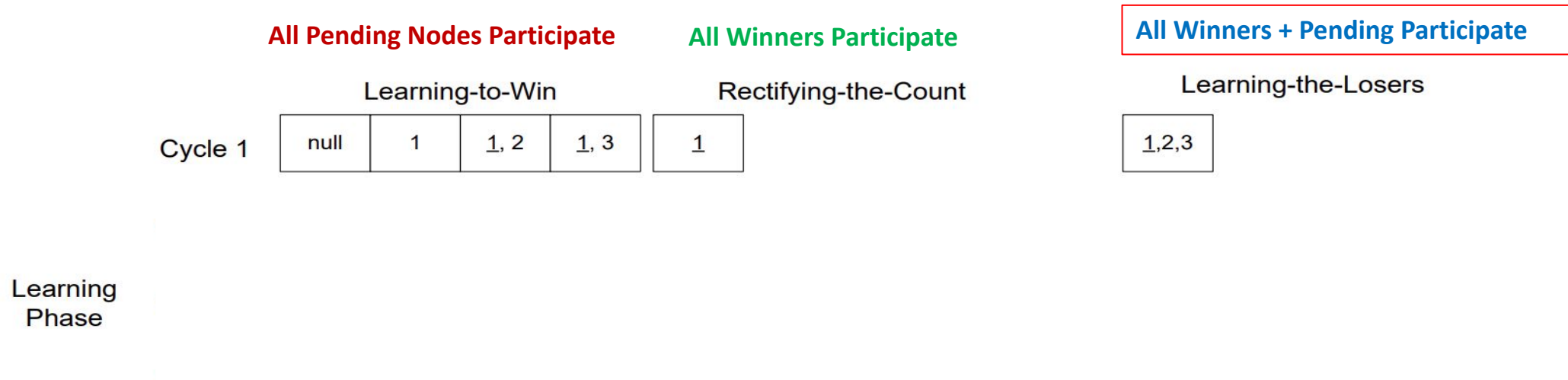


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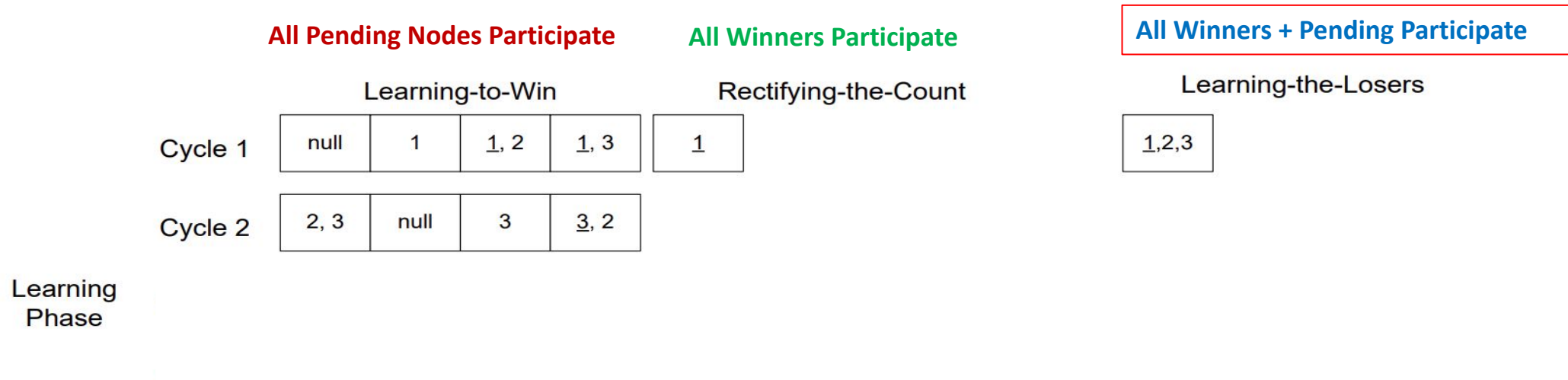


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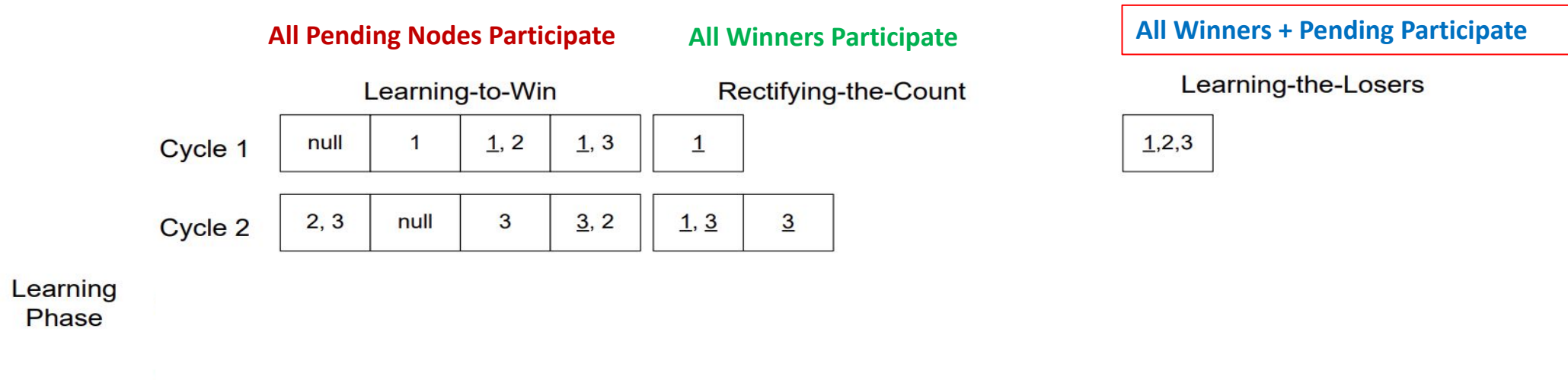


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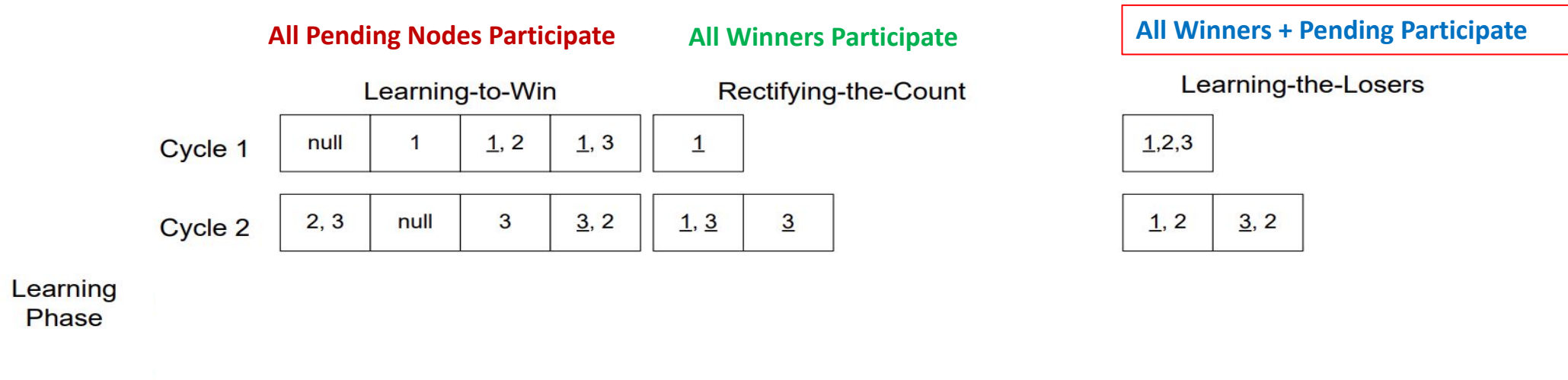


Fig. 2. Illustration of PC protocol when N is known. ($N = 3, K = 4, N^{max} = 4$)

PC: Learning Phase (while Number of Nodes are **known to All**)

		All Pending Nodes Participate				All Winners Participate		All Winners + Pending Participate		
		Learning-to-Win				Rectifying-the-Count		Learning-the-Losers		
Learning Phase	Cycle 1	null	1	1, 2	1, 3	1		1,2,3		
	Cycle 2	2, 3	null	3	<u>3</u> , 2	<u>1</u> , <u>3</u>	<u>3</u>	<u>1</u> , 2	<u>3</u> , 2	
	Cycle 3	null	null	null	null	1	<u>3</u>	1, 2	<u>3</u> , 2	null

Fig. 2. Illustration of PC protocol when N is known. ($N = 3, K = 4, N^{max} = 4$)

PC: Learning Phase (while Number of Nodes are **known to All**)

All Pending Nodes Participate					All Winners Participate			All Winners + Pending Participate				
Learning Phase		Learning-to-Win			Rectifying-the-Count			Learning-the-Losers				
	Cycle 1	null	1	<u>1</u> , 2	<u>1</u> , 3	1			1,2,3			
	Cycle 2	2, 3	null	3	<u>3</u> , 2	<u>1</u> , <u>3</u>	<u>3</u>		<u>1</u> , 2	<u>3</u> , 2		
	Cycle 3	null	null	null	null	<u>1</u>	<u>3</u>	null	<u>1</u> , 2	<u>3</u> , 2	null	
	Cycle 4	null	null	null	null	<u>1</u>	<u>3</u>	null	null	<u>1</u> , 2	<u>3</u> , 2	null

Fig. 2. Illustration of PC protocol when N is known. ($N = 3, K = 4, N^{max} = 4$)

PC: Learning Phase (while Number of Nodes are **known to All**)

Learning Phase	All Pending Nodes Participate				All Winners Participate		All Winners + Pending Participate				
	Learning-to-Win				Rectifying-the-Count		Learning-the-Losers				
	Cycle 1	null	1	1, 2	1, 3	1	1,2,3				
	Cycle 2	2, 3	null	3	<u>3</u> , 2	<u>1</u> , <u>3</u> <u>3</u>	<u>1</u> , 2 <u>3</u> , 2				
	Cycle 3	null	null	null	null	<u>1</u> <u>3</u> null	<u>1</u> , 2 <u>3</u> , 2 null				
	Cycle 4	null	null	null	null	<u>1</u> <u>3</u> null null	<u>1</u> , 2 <u>3</u> , 2 null null				
	Cycle 5	null	2	<u>2</u>	<u>2</u>	<u>1</u> , <u>2</u> <u>3</u> , <u>2</u> <u>2</u> null null	<u>1</u> <u>3</u> <u>2</u> null null				

Fig. 2. Illustration of PC protocol when N is known. ($N = 3, K = 4, N^{max} = 4$)

PC: Learning Phase (while Number of Nodes are **known to All**)

All Pending Nodes Participate				All Winners Participate		All Winners + Pending Participate													
Learning Phase		Learning-to-Win			Rectifying-the-Count				Learning-the-Losers										
	Cycle 1	null	1	1, 2	1, 3	1				1,2,3									
	Cycle 2	2, 3	null	3	3, 2	1, 3	3				1, 2	3, 2							
	Cycle 3	null	null	null	null	1	3	null				1, 2	3, 2	null					
	Cycle 4	null	null	null	null	1	3	null	null				1, 2	3, 2	null	null			
Cycle 5	null	2	2	2	1, 2	3, 2	2	null	null				1	3	2	null	null		
Transmission Phase		1	3	2	1	3	2	...											

Fig. 2. Illustration of PC protocol when N is known. ($N = 3, K = 4, N^{max} = 4$)

Paper Summary: Advantage

- Fully Distributed
→ Works without any central controller or coordination messages between devices.
- Highly Efficient
→ No collisions in Transmission Phase and minimal wasted time.
- No Idle Time Before Transmission
→ Unlike 802.11 which waits during backoff, it doesn't waste time sensing an idle channel.
- Convergence to steady state very fast
- Doesn't require Special PHY (OFDM) Layer
- Doesn't require Additional Antenna at each Nodes

Perfect Coordination (PC) protocol: Weaknesses

- PC requires all the nodes being **tightly Time Synchronized**
- **Need a dedicated Learning Phase and the Learning Phase is Complex**
- **Not efficient for Highly Dynamic Networks (handling newly arrived and left nodes demands repeating learning phase too frequently)**

Solution Approach -3.1

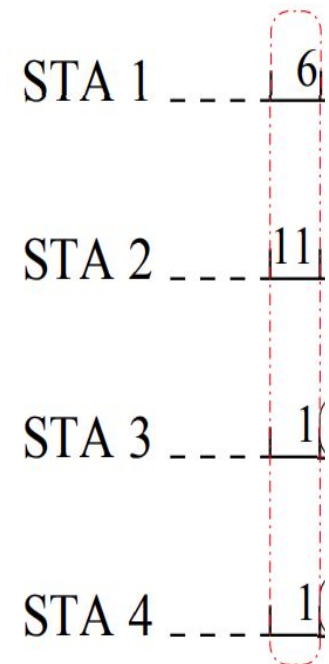
A Distributed approach for gradually achieving collision-free Channel Access using Deterministic backoff after successful transmissions

Paper Title: **Deterministic Backoff** :: A High Efficiency MAC Protocol for WLANs: Providing Fairness in Dense Scenarios

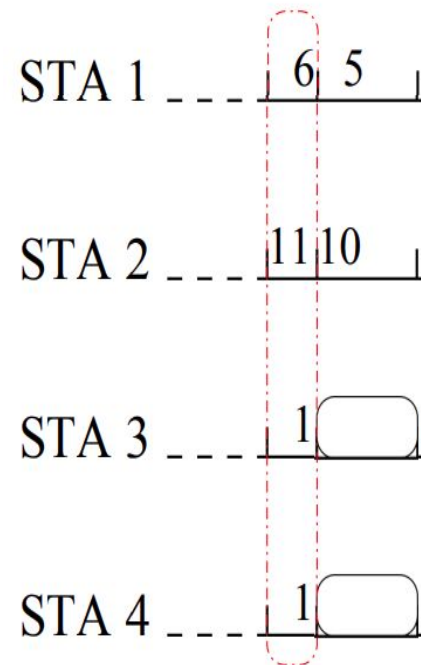
Publisher: *IEEE/ACM Transactions on Networking* (2016), *Communications Workshops (ICC)*, 2013

- (CSMA/ECA) can create a collision-free schedule in a fully decentralized manner using a **deterministic backoff (Pre-Determined)** after successful transmissions.
- Hysteresis and Fair Share are two extensions of CSMA/ECA to support a large number of contenders in a collision-free schedule.
- Coexistence and backwards compatibility with CSMA/CA nodes under different traffic conditions.

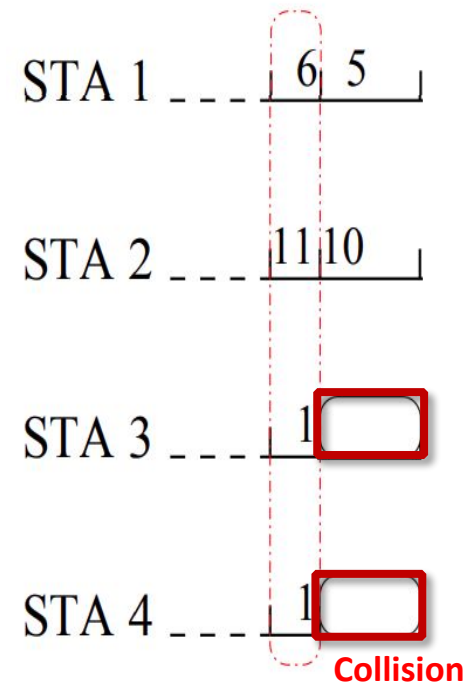
Deterministic Backoff : Basic Concept



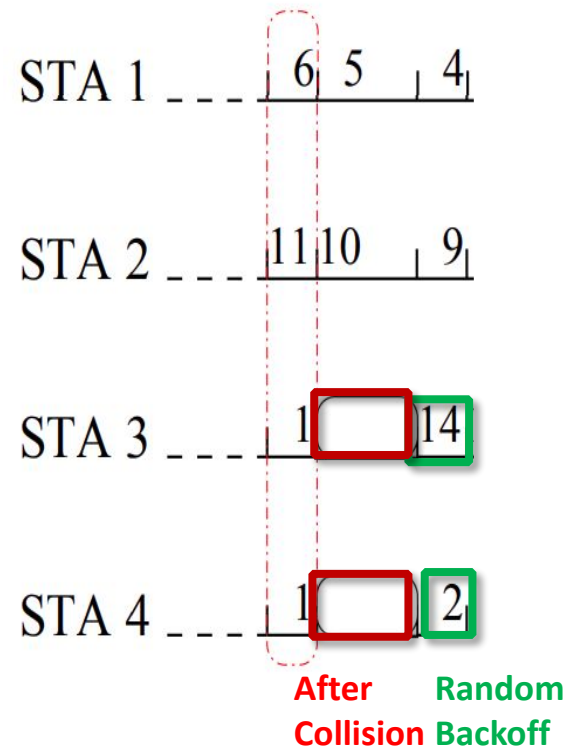
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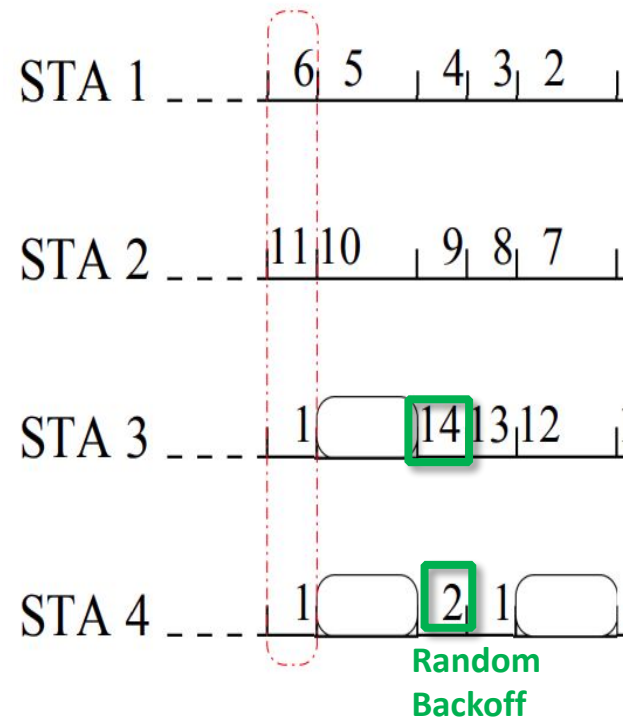
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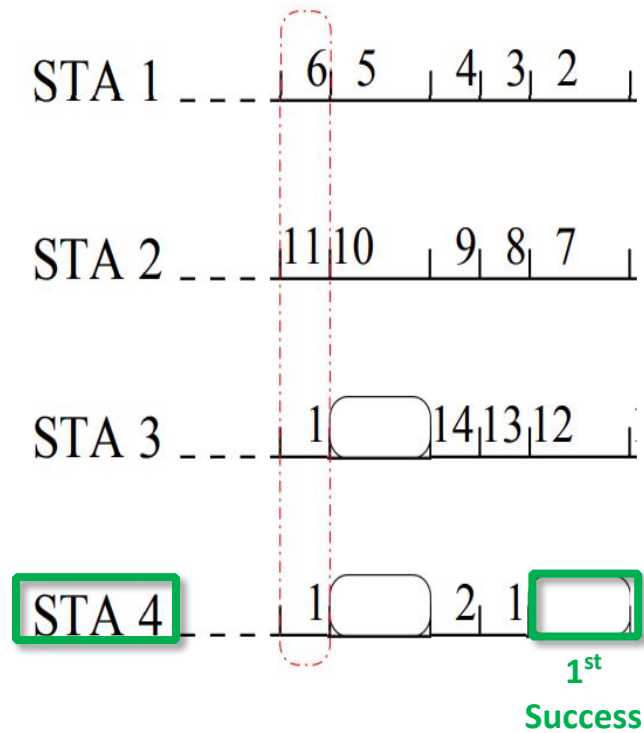
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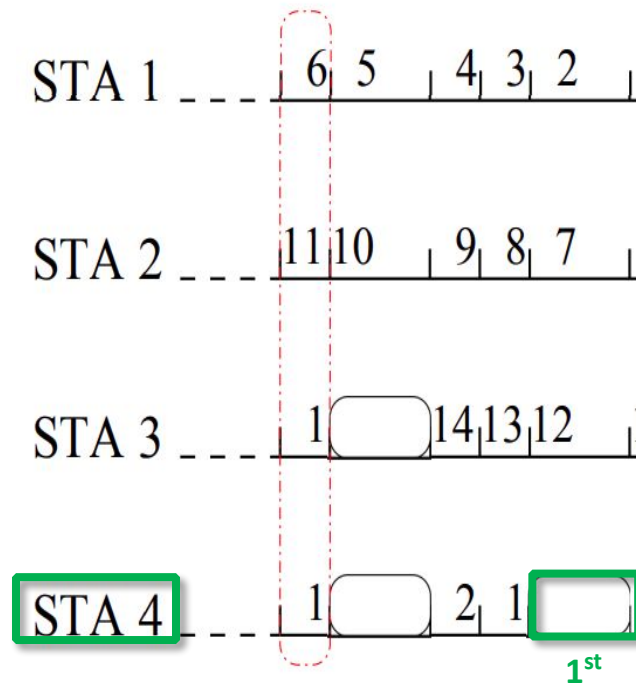
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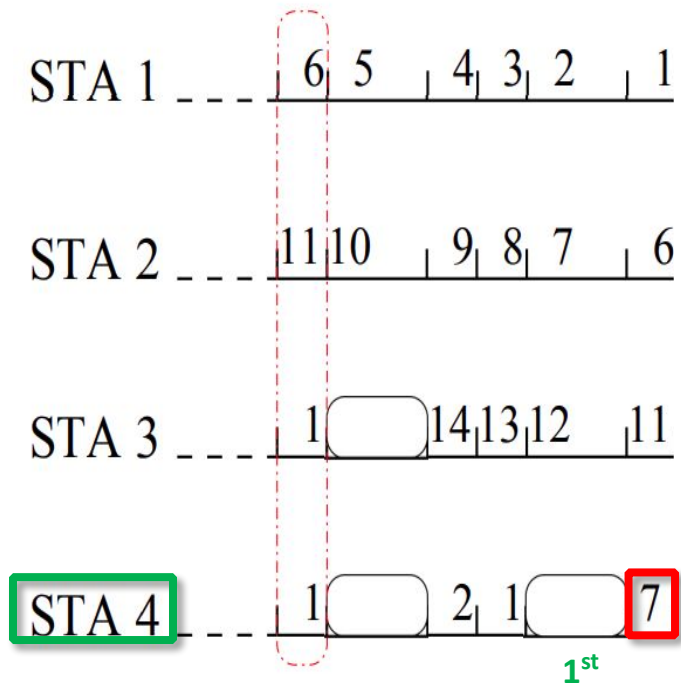
Deterministic Backoff : Basic Concept



Success

Deterministic Backoff
Let, 7

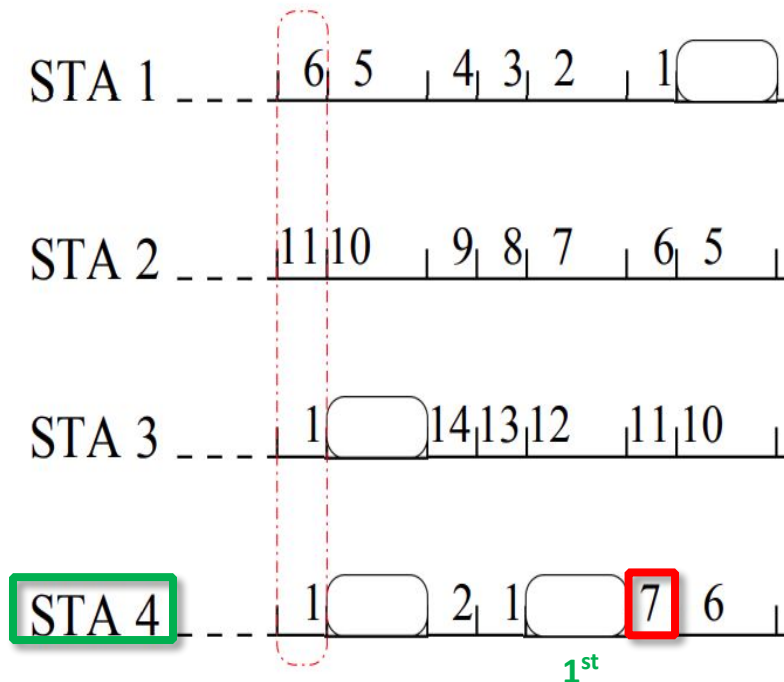
Deterministic Backoff : Basic Concept



Success

Deterministic Backoff
Let, 7

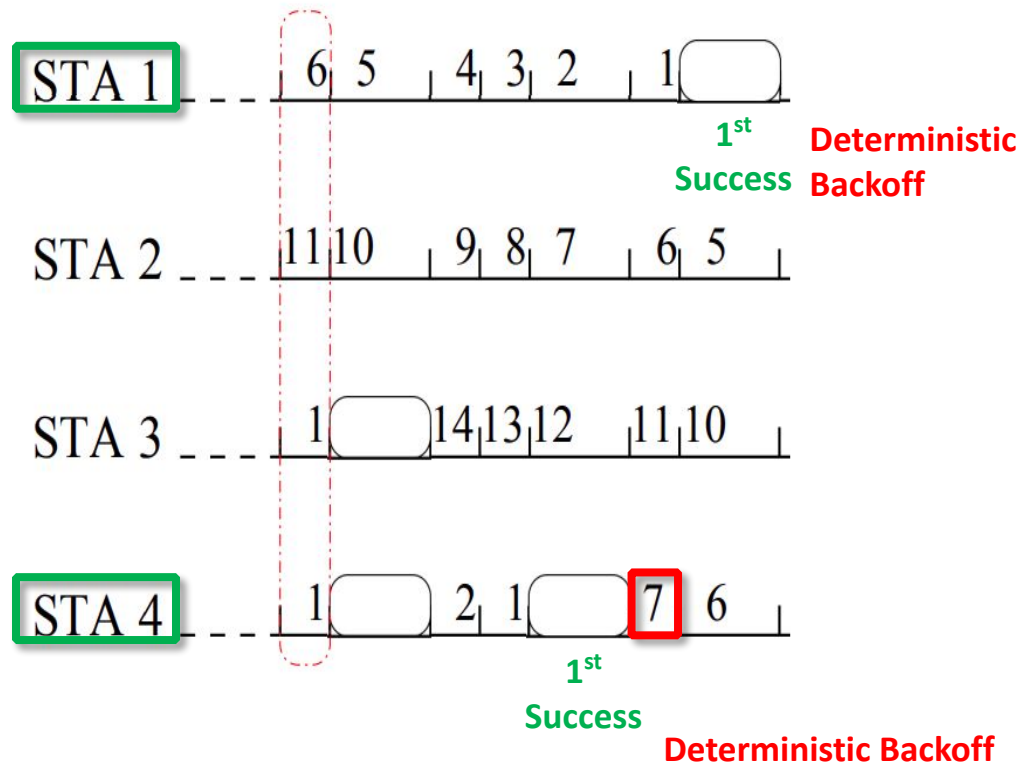
Deterministic Backoff : Basic Concept



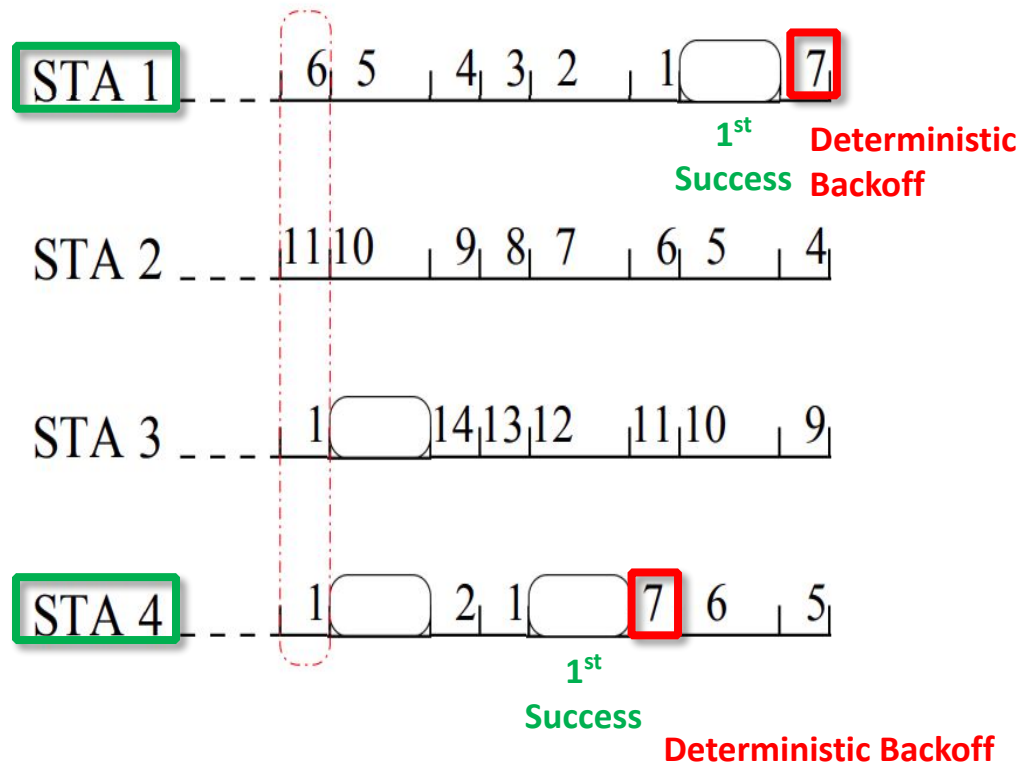
Success

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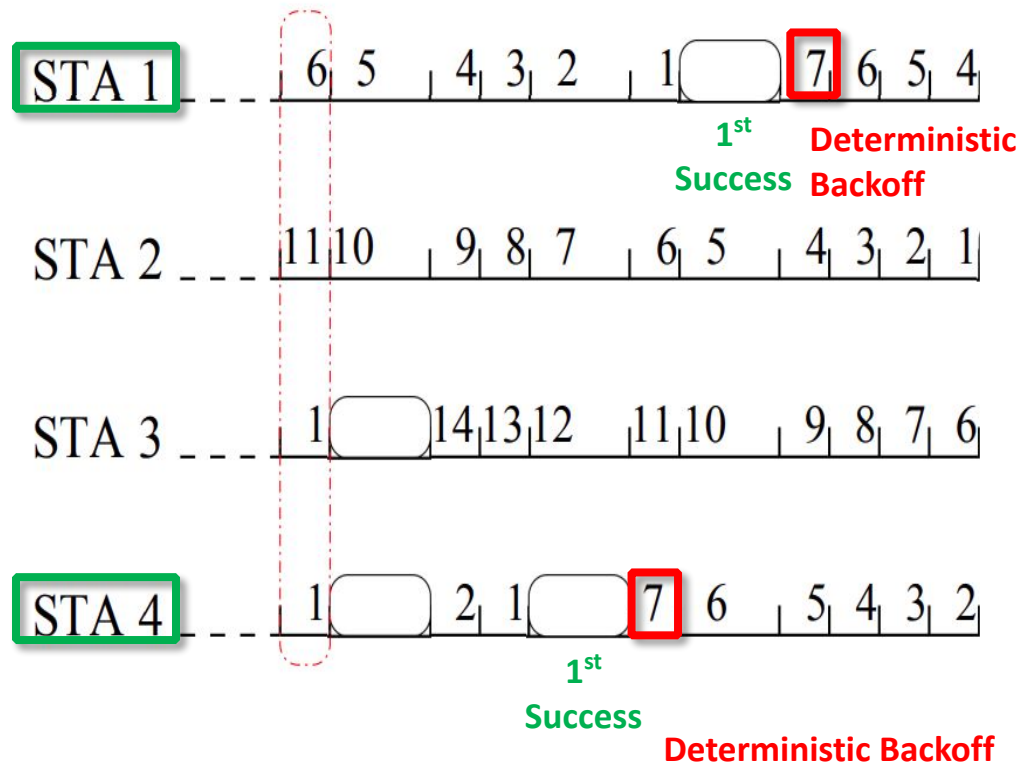
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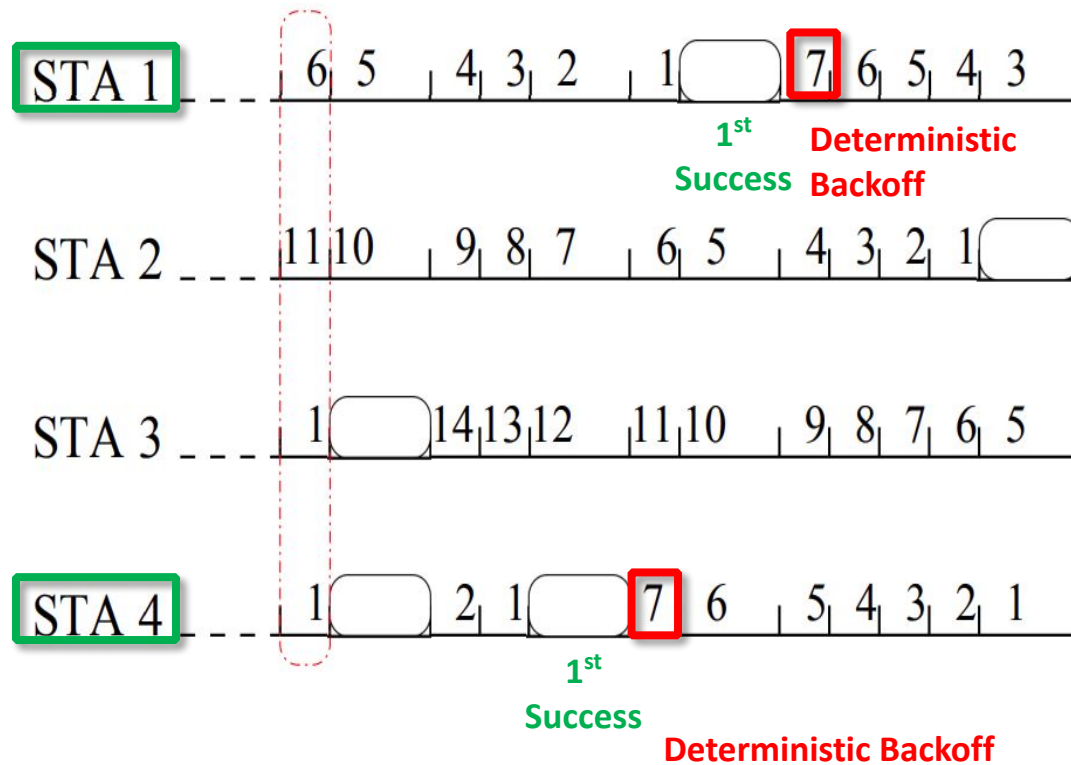
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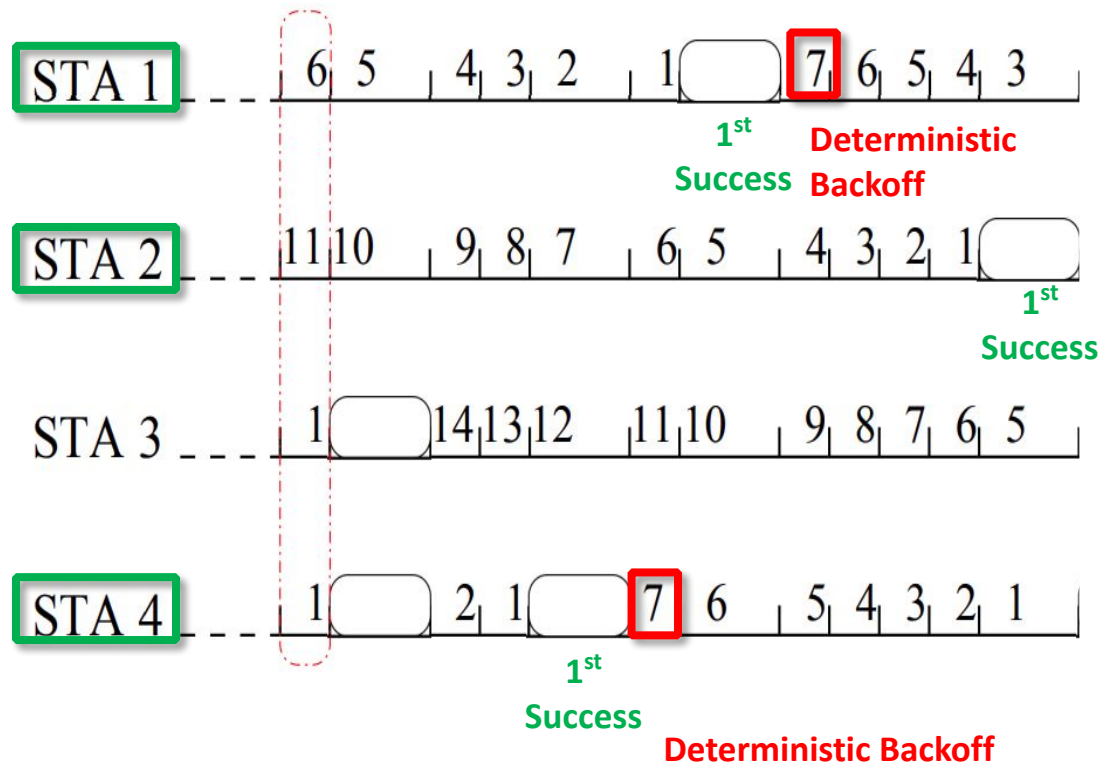
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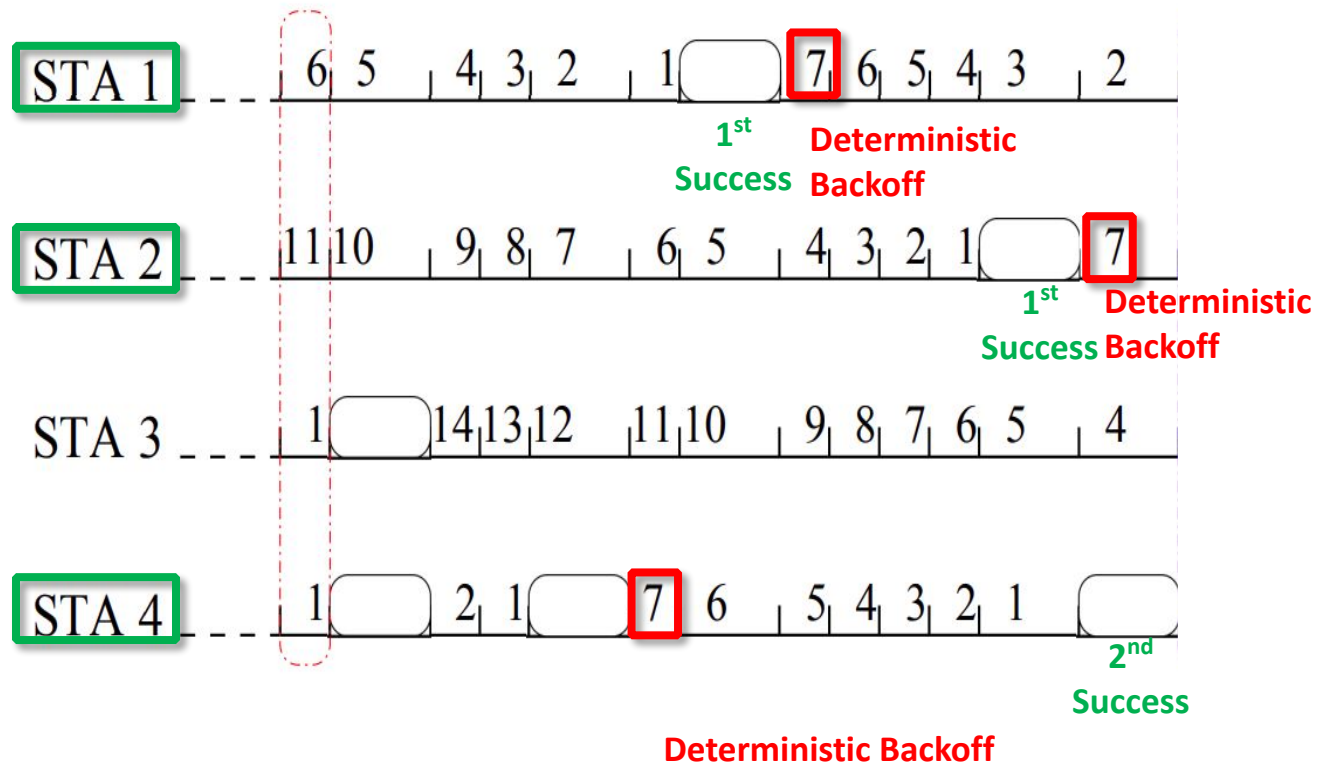
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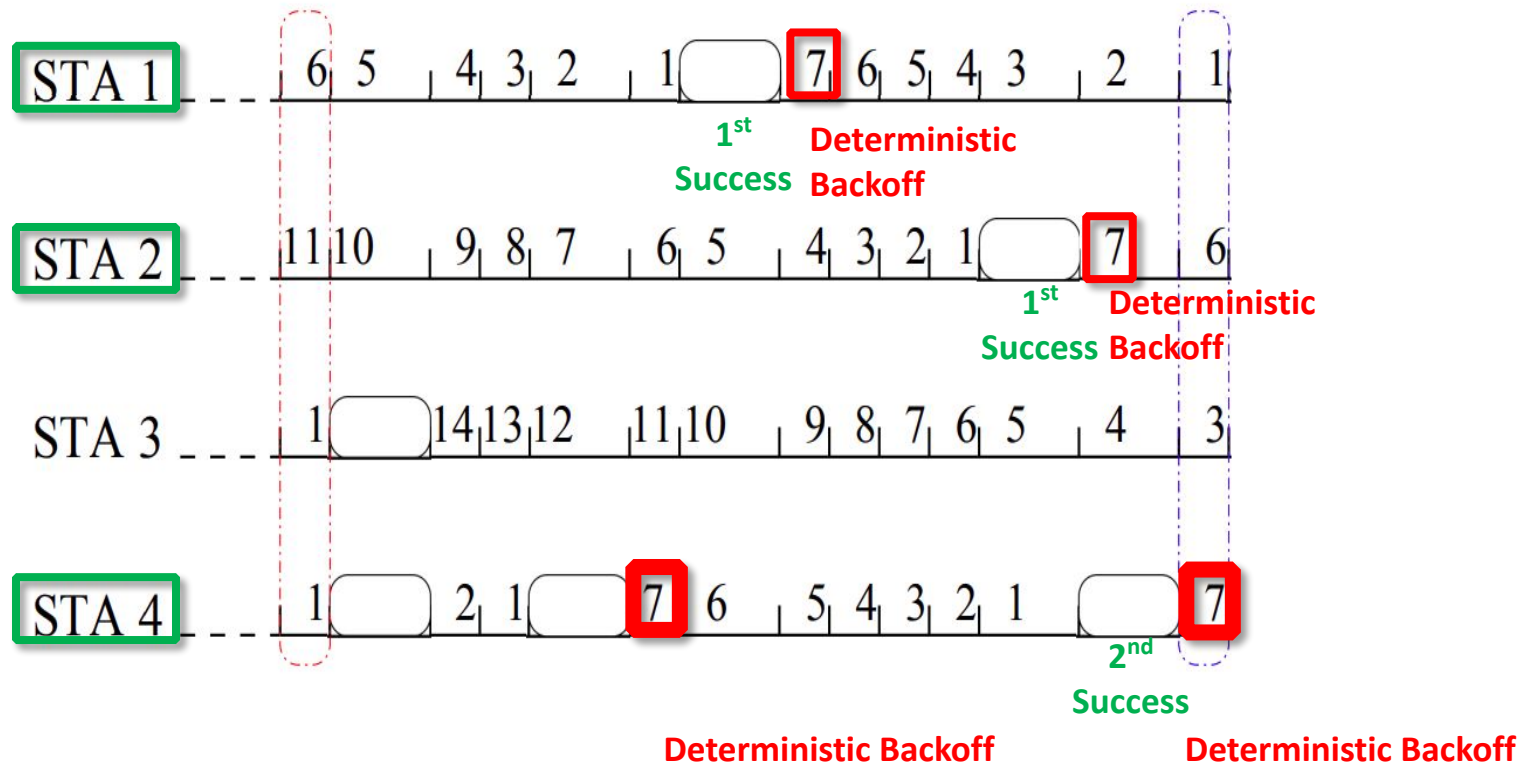
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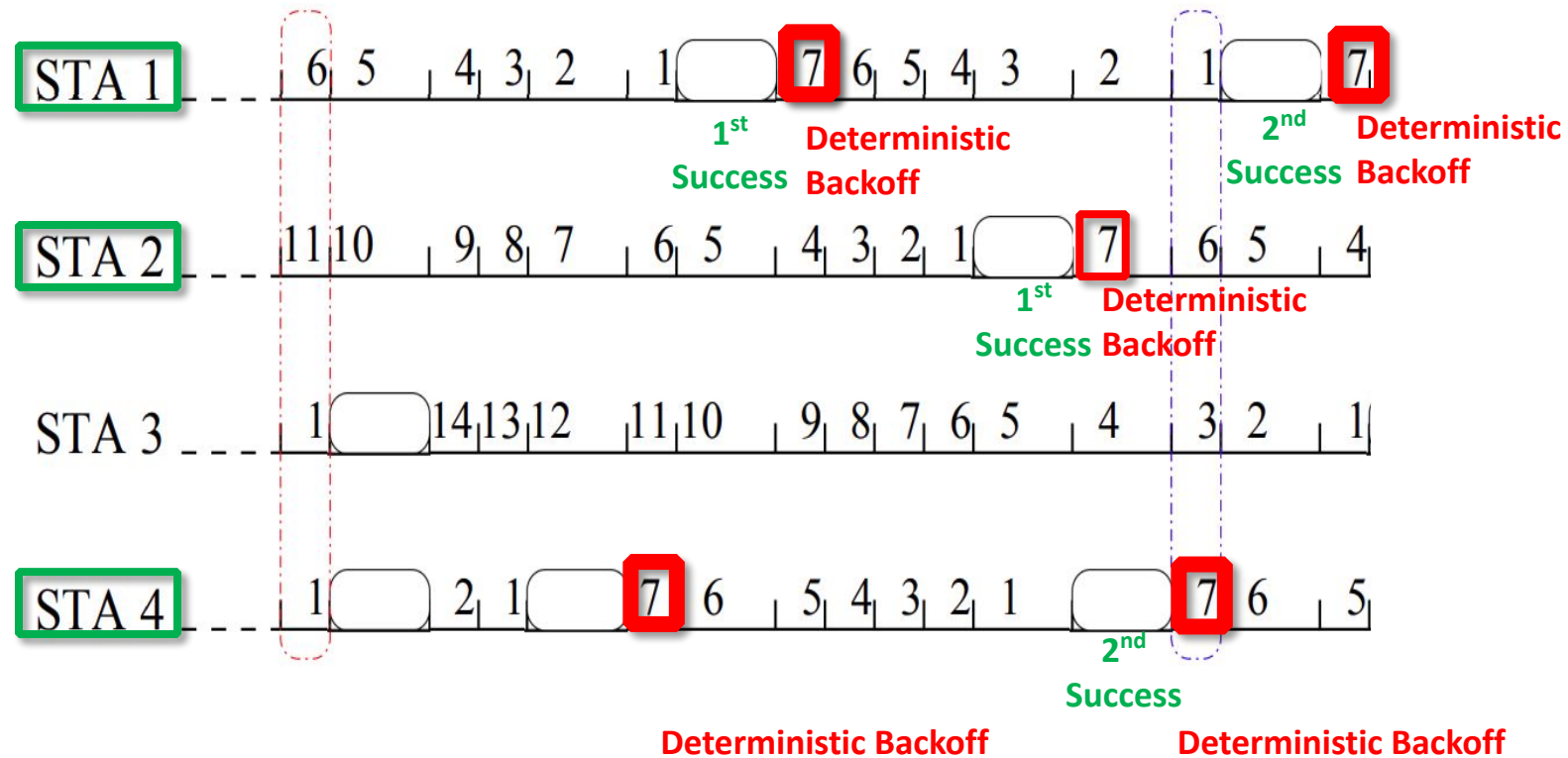
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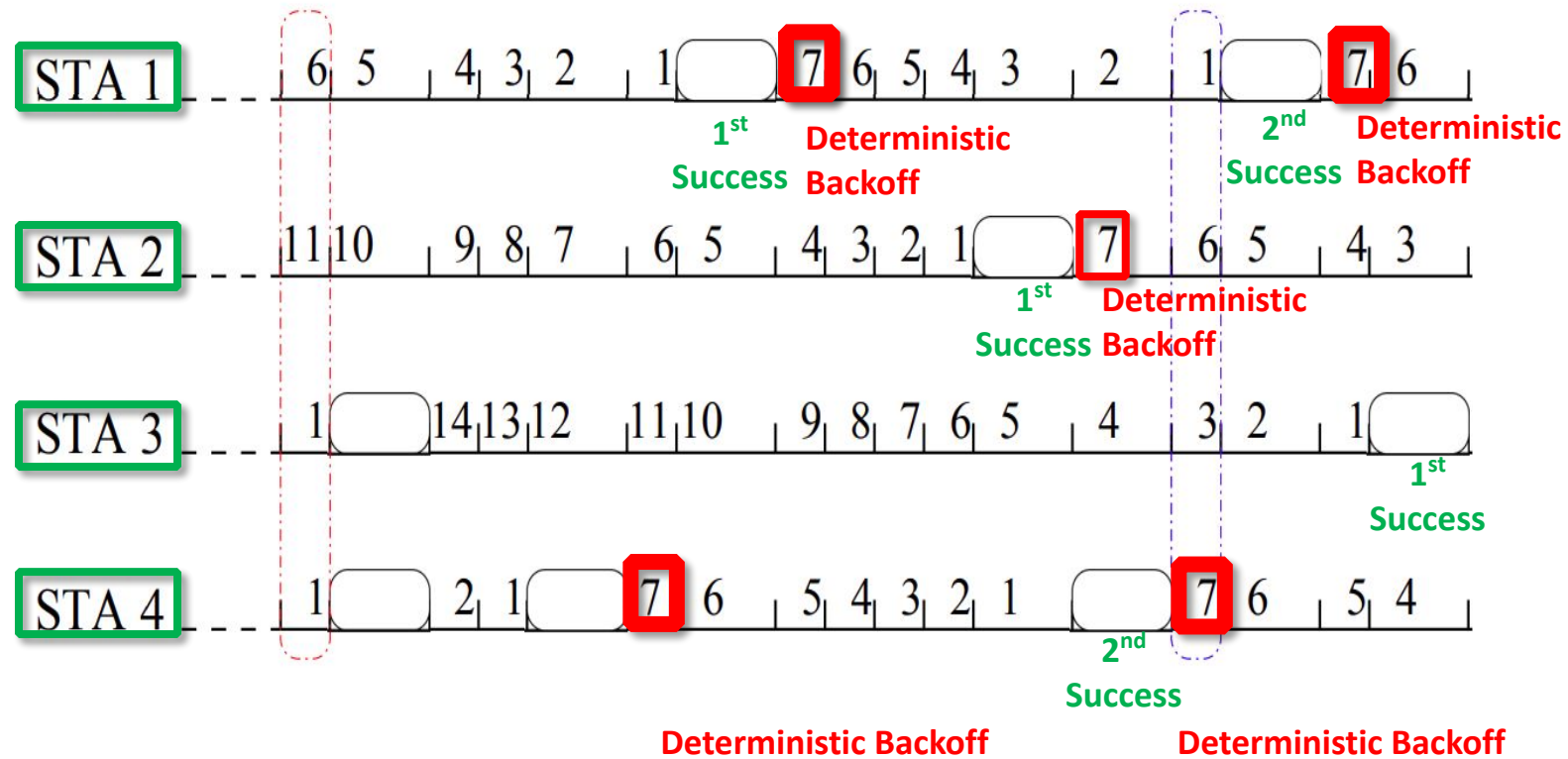
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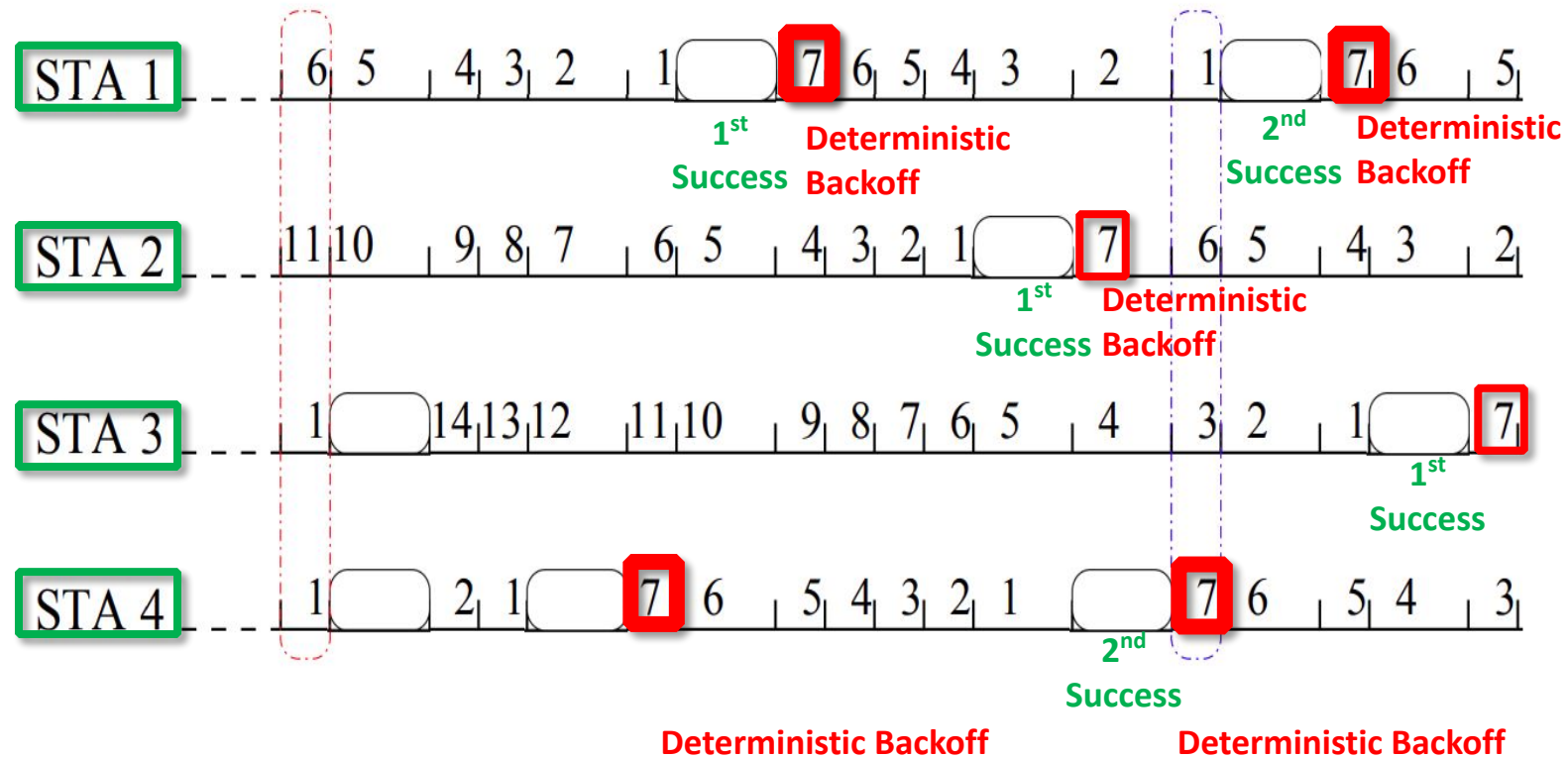
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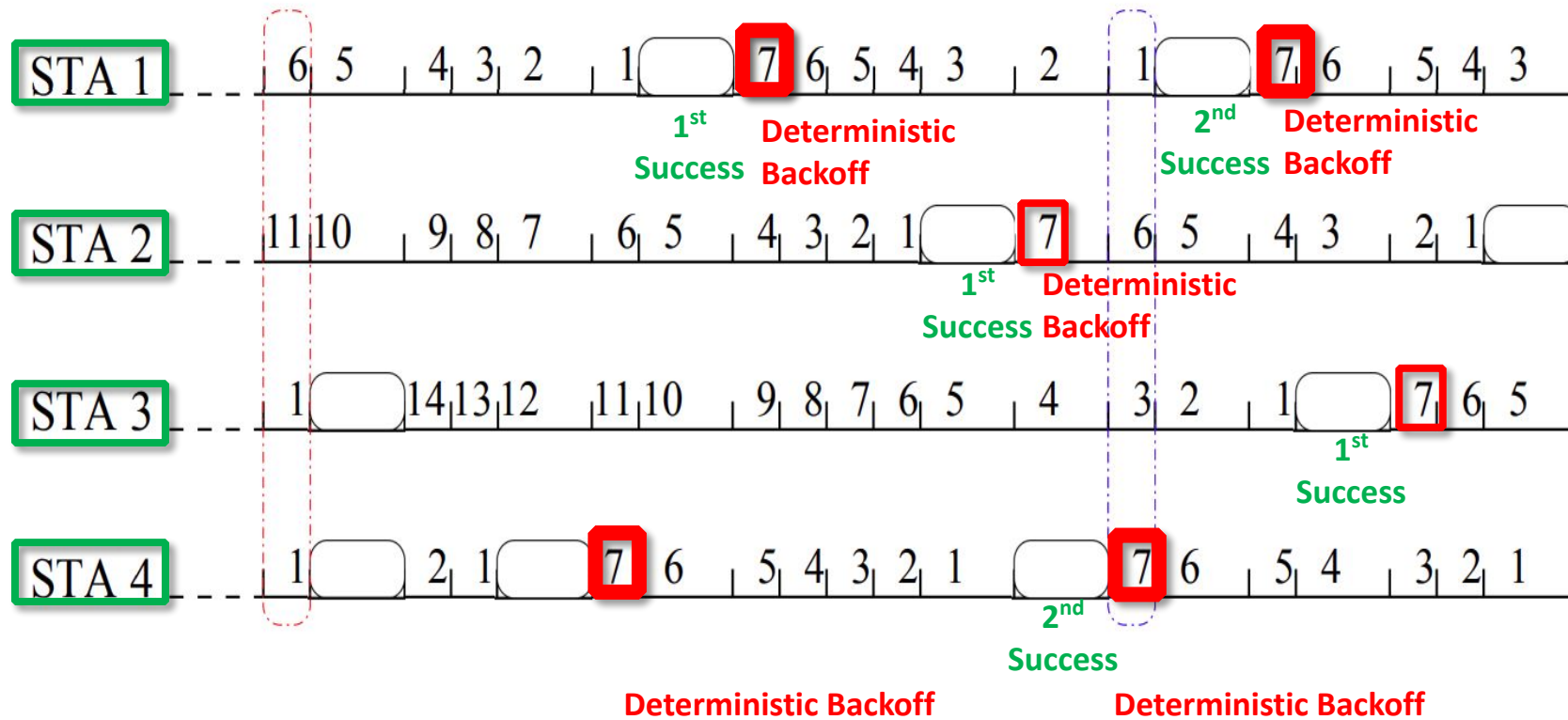
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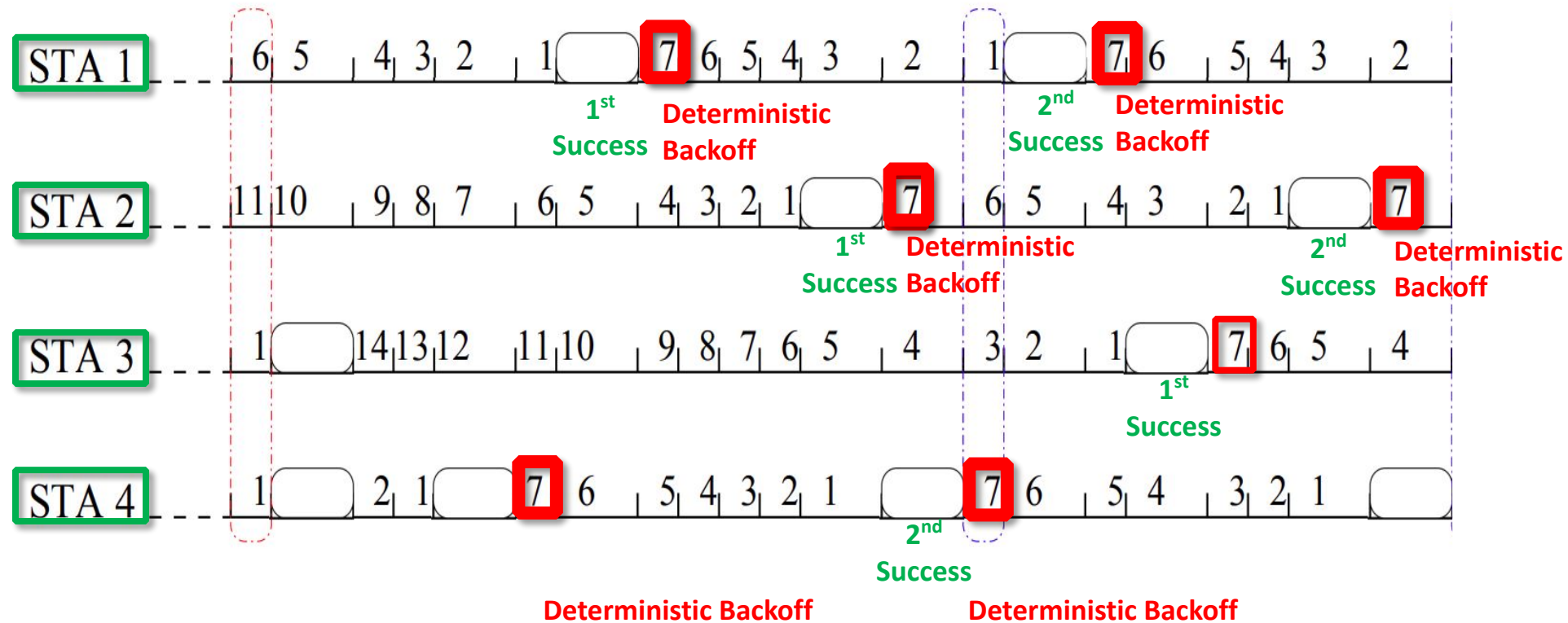
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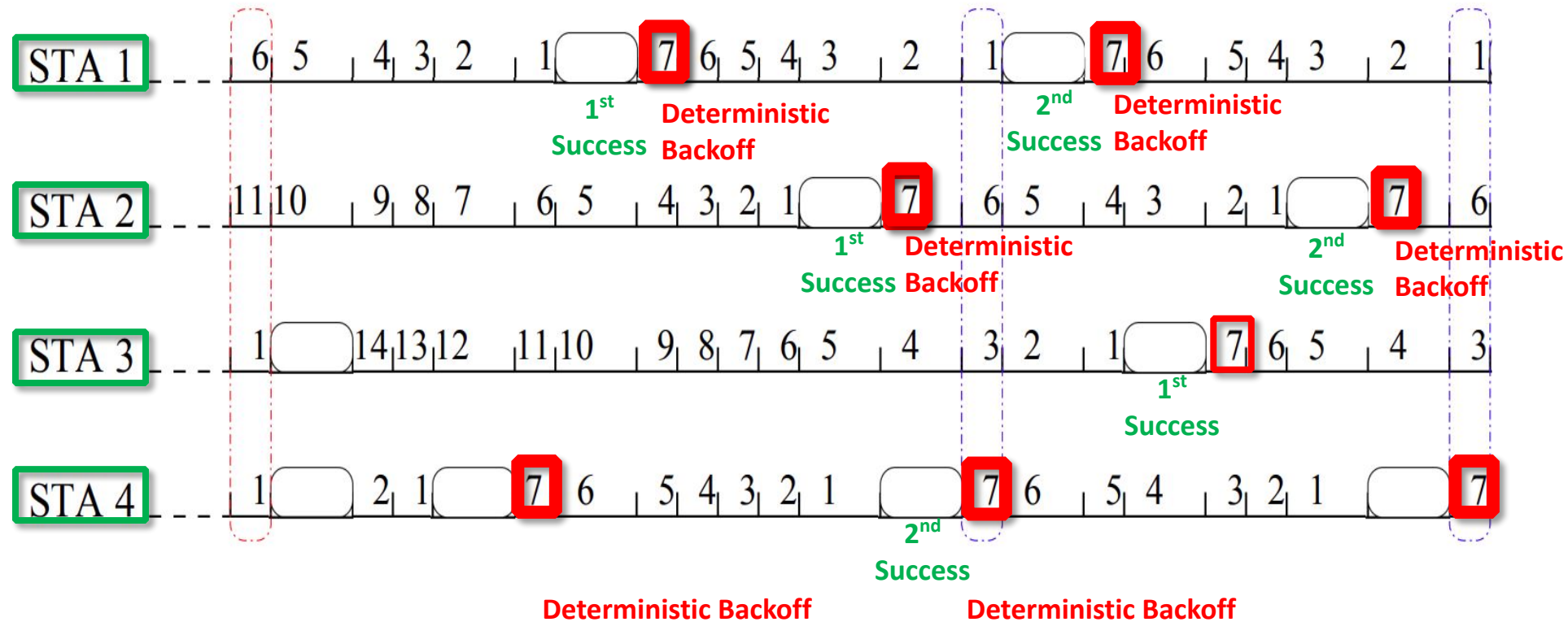
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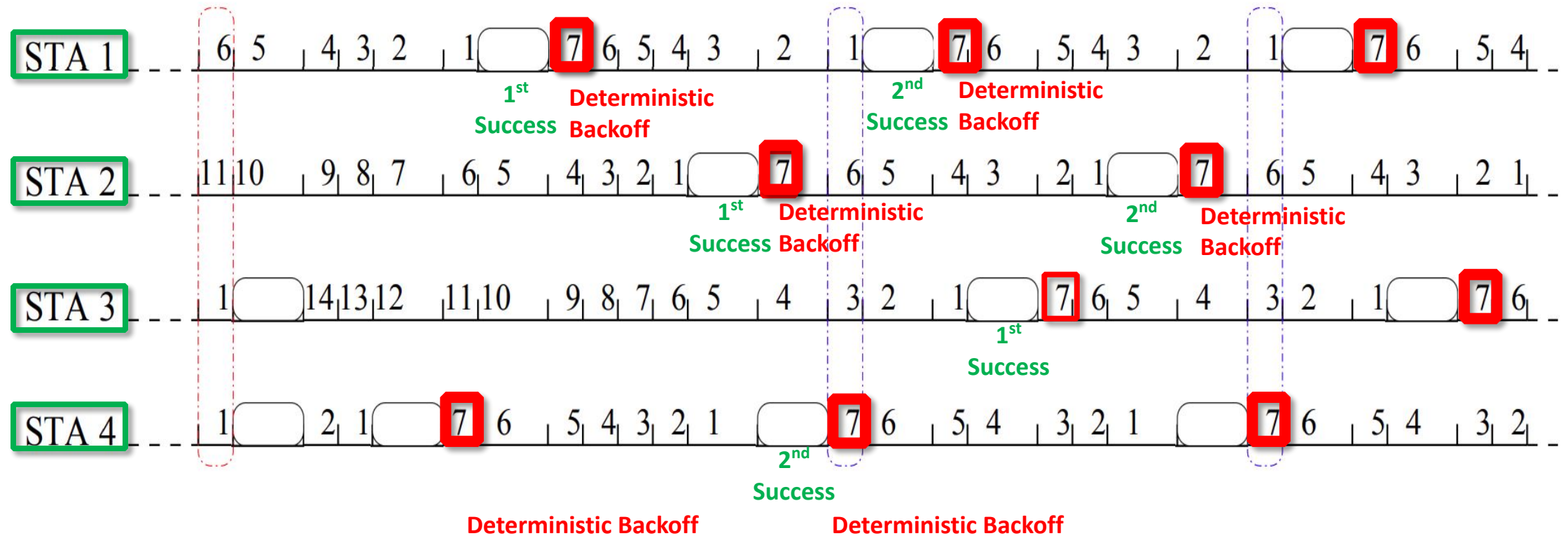
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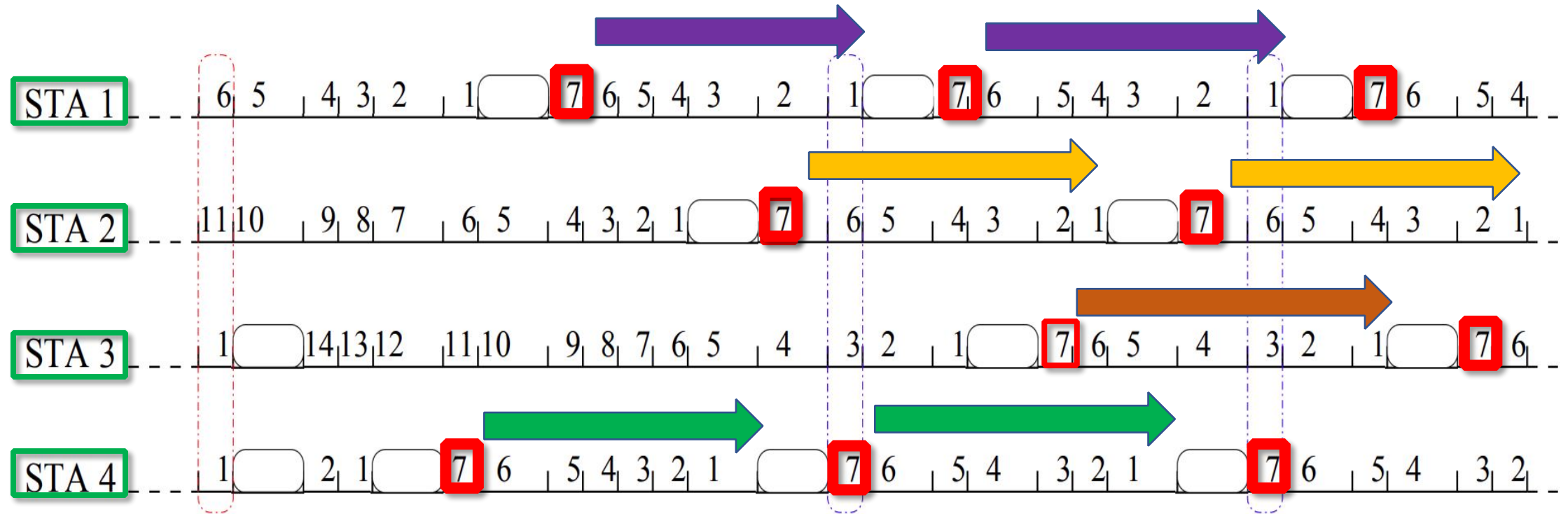
Deterministic Backoff : Basic Concept



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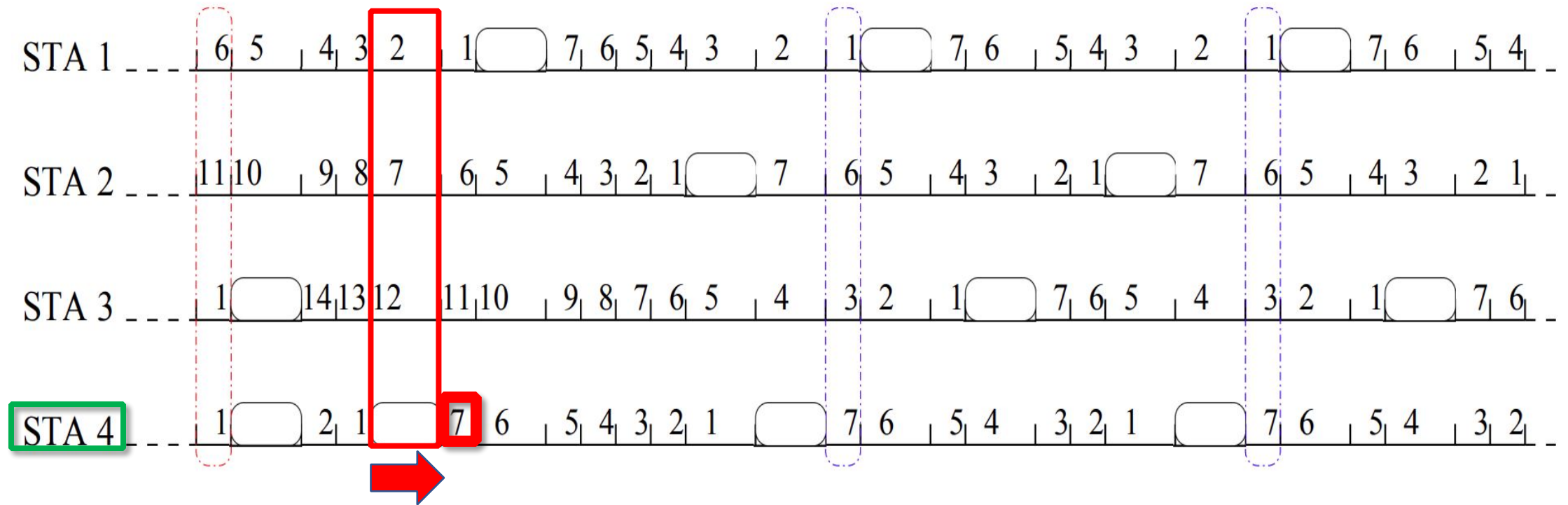


Deterministic Backoff : Basic Concept



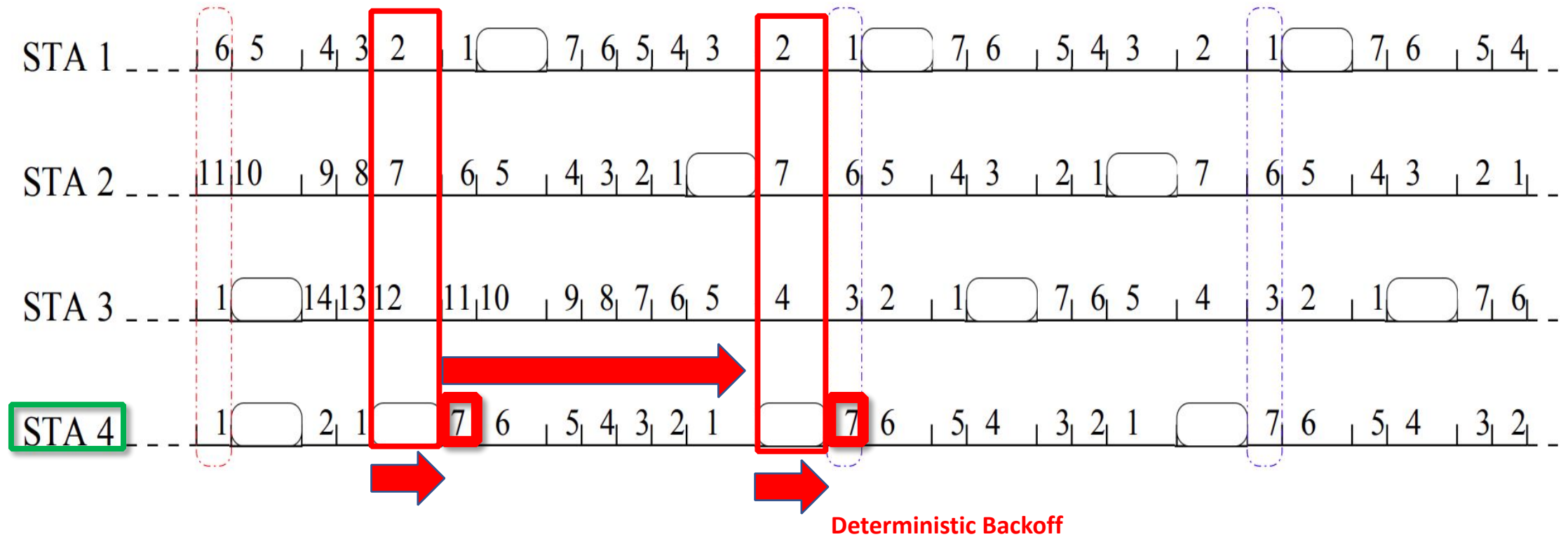
Deterministic Backoff

Deterministic Backoff : Basic Concept

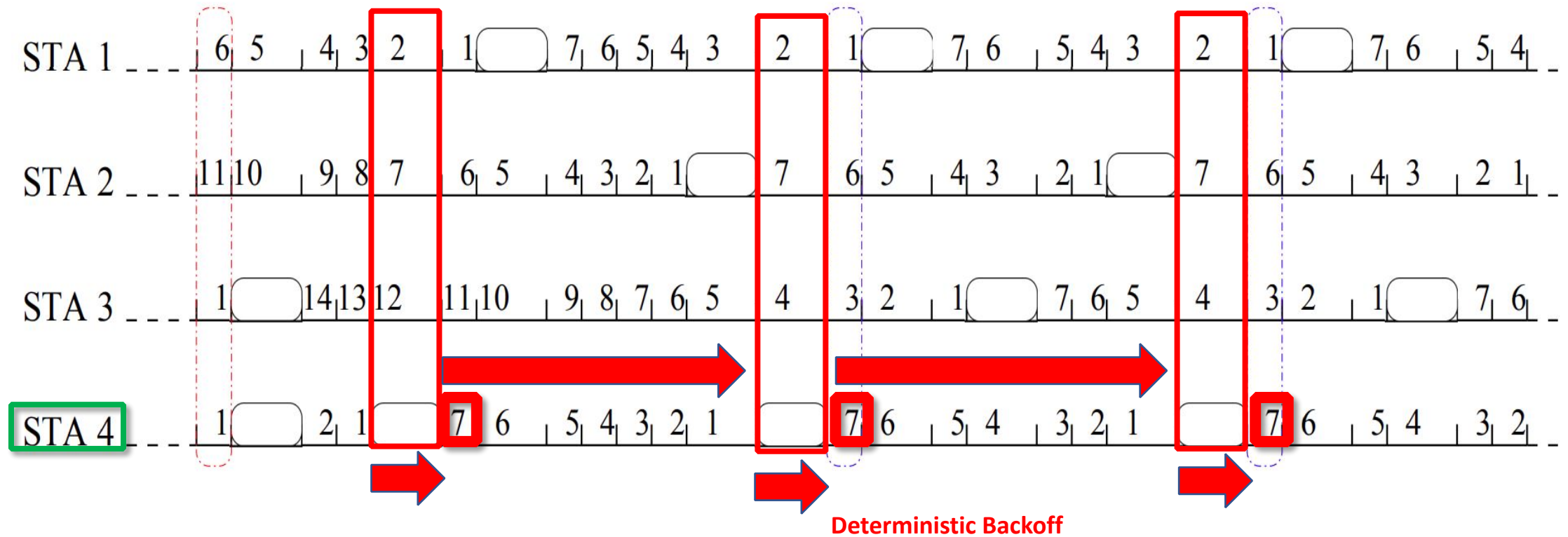


Deterministic Backoff

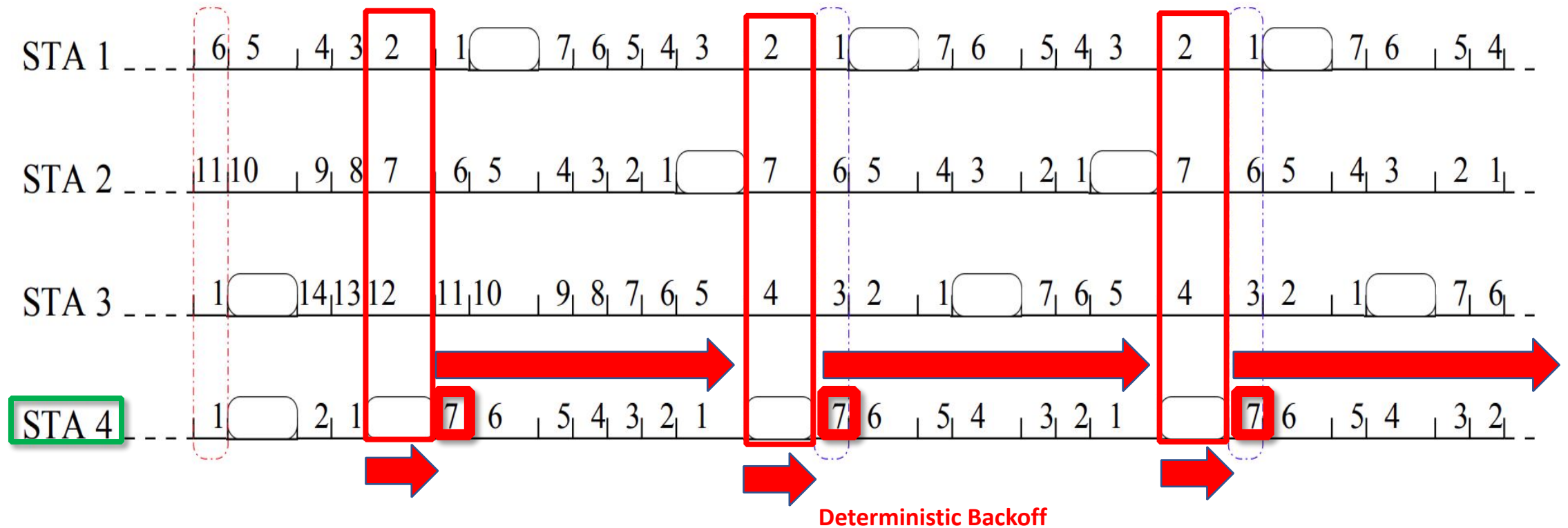
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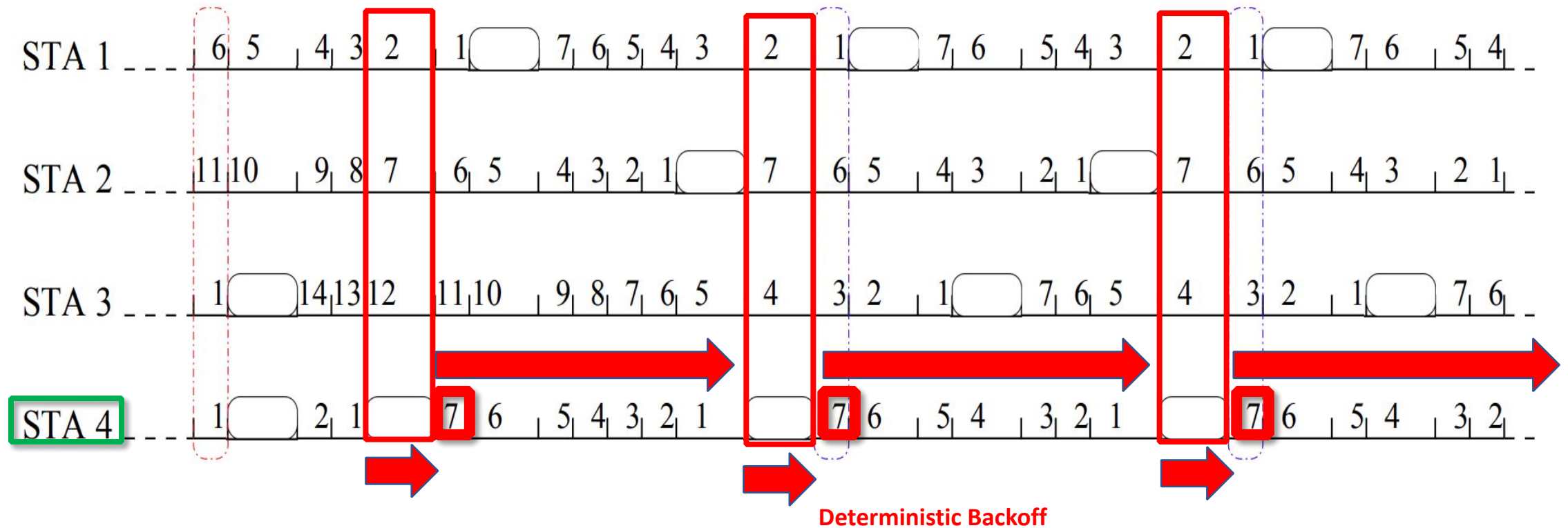
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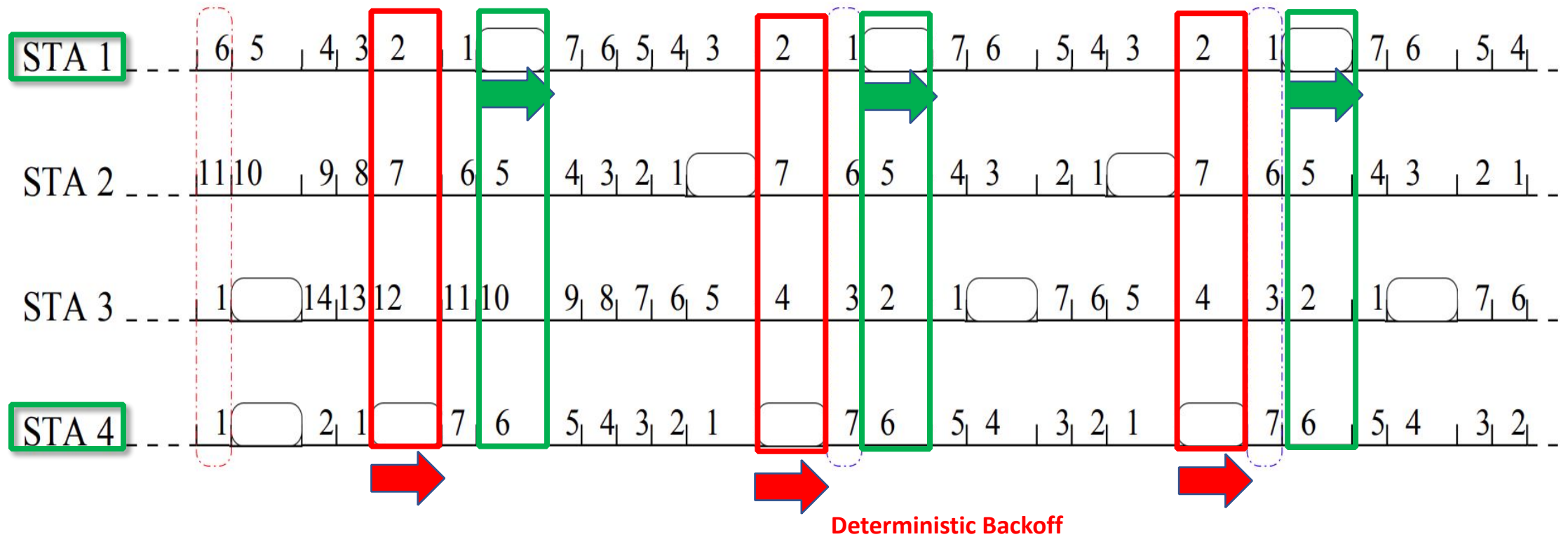
Deterministic Backoff : Basic Concept



Deterministic Backoff : Basic Concept



Deterministic Backoff : Basic Concept



Deterministic Backoff : Basic Concept

```
1 while the device is on do
2    $r \leftarrow 0$  ;  $k \leftarrow 0$ ;
3    $B \leftarrow \mathcal{U}[0, 2^k CW_{\min} - 1]$ ;
4   while there is a packet to transmit do
5     repeat
6       while  $B > 0$  do
7         wait 1 slot;
8          $B \leftarrow B - 1$ ;
9         Attempt transmission of 1 packet;
10        if collision then
11           $r \leftarrow r + 1$ ;
12           $k \leftarrow \min(k + 1, m)$ ;
13           $B \leftarrow \mathcal{U}[0, 2^k CW_{\min} - 1]$ ;
14        until ( $r = R$ ) or (success);
15         $r \leftarrow 0$ ;
16         $k \leftarrow 0$ ;
17        if success then
18           $B_d \leftarrow \lceil 2^k CW_{\min} / 2 \rceil - 1$ ;
19           $B \leftarrow B_d$ ;
20        else
21          Discard packet;
22           $B \leftarrow \mathcal{U}[0, 2^k CW_{\min} - 1]$ ;
23    Wait until there is a packet to transmit;
```

Let, $CW_{\min} = 16$
Then $B_d = 7$

Deterministic Backoff :: Strength

- **Its very Simple**
- **Doesn't require Special PHY (OFDM) Layer**
- **Doesn't require Additional Antenna at each Nodes**
- **Doesn't require all the nodes being tightly Time Synchronized**
- **Doesn't require any dedicated and Complex Learning Phase**
- **Schedule Preparation is very simple**
- **Converge to steady state is not very slow**
- **Scalable (Support a large number of contenders in a collision-free schedule)**
- **Handling the New node arrival and departure is trivial**
- **Coexistence and backwards compatibility with CSMA/CA**
- **Once the network converges to steady state, there will be no collisions and channel wastage will be minimum**

Deterministic Backoff : Weaknesses

- Throughput efficiency may not reach optimal levels, but it is still significantly better than that of random-access mechanisms.
- Not highly efficient in highly dynamic networks, as newly arrived nodes can disrupt the network's steady state and cause instability again.

Solution Approach -3.2

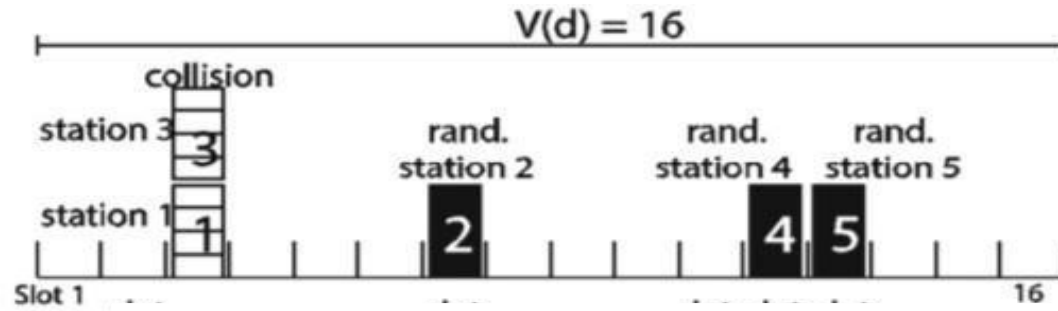
A Distributed approach for gradually achieving collision-free Channel Access using Deterministic backoff and Beacon Transmission

Paper Title: A Beacon-Based Collision-Free Channel Access Scheme for IEEE 802.11 WLANs

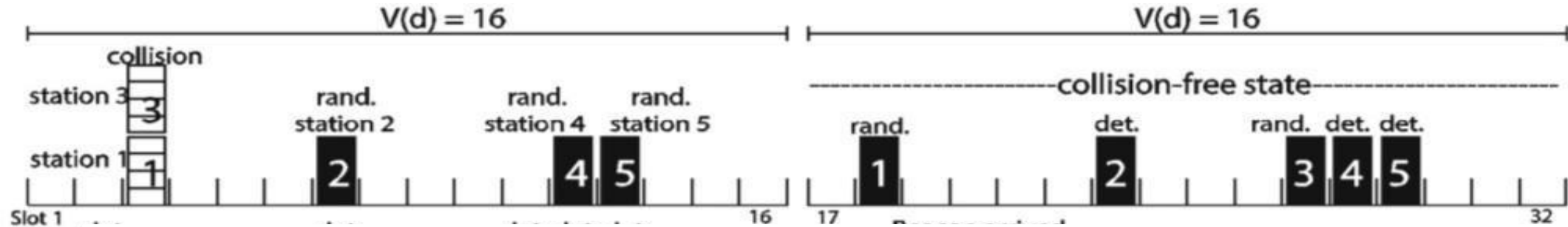
Publisher: *Wireless personal communications* (2014)

- The proposed scheme makes **use of beacon frames sent periodically by access point, lets stations enter the collision-free state and reduces the number of idle slots** regardless of the number of stations and their traffic load (saturated or unsaturated) on the medium.

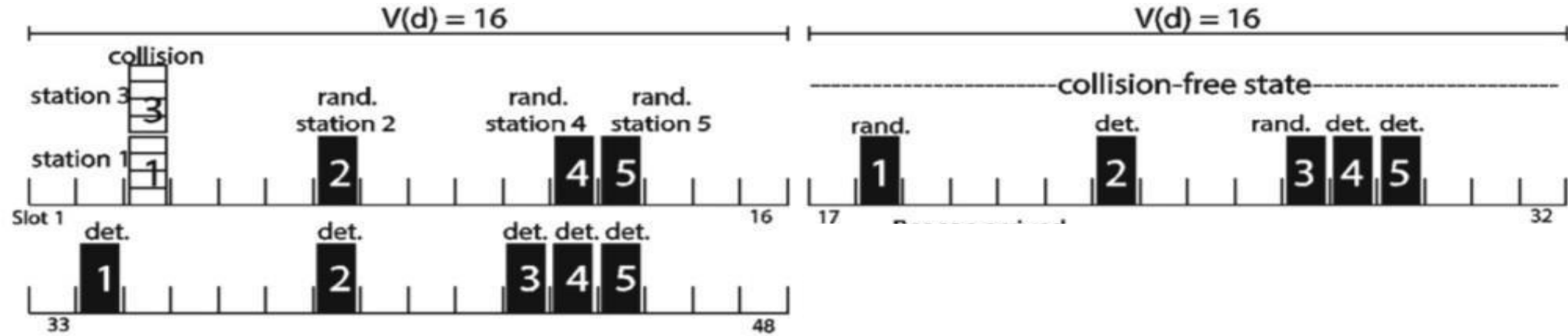
Beacon assisted Deterministic Backoff : Basic Concept



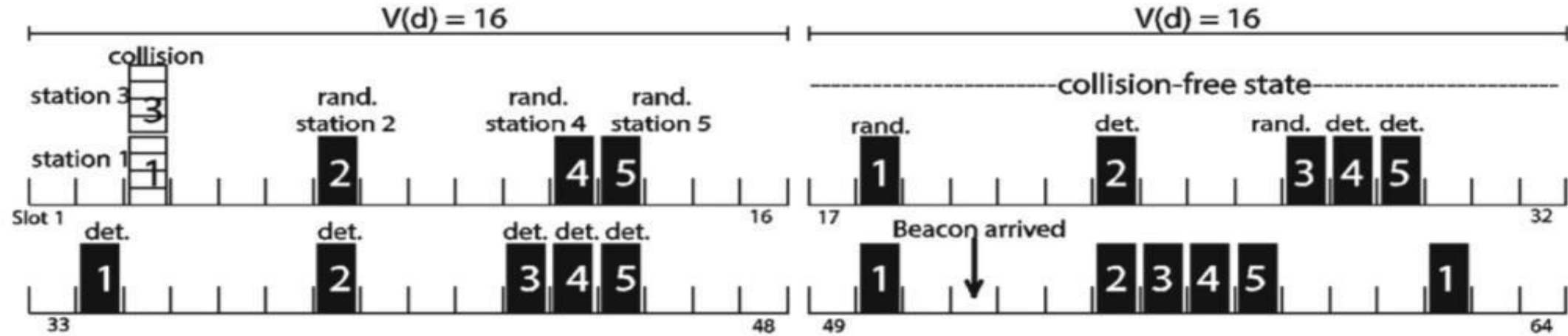
Beacon assisted Deterministic Backoff : Basic Concept



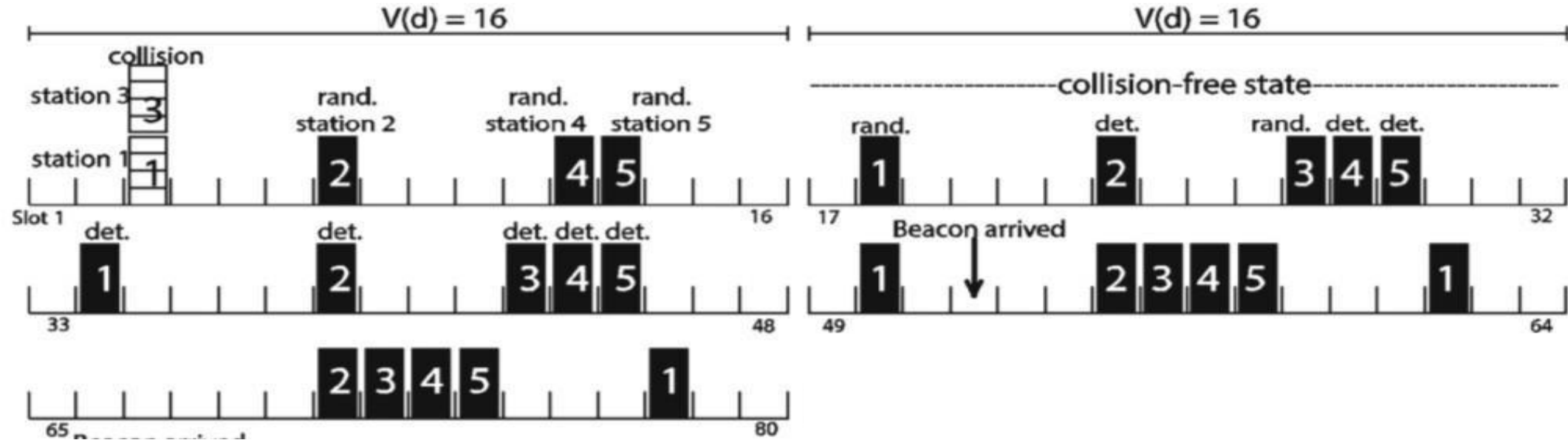
Beacon assisted Deterministic Backoff : Basic Concept



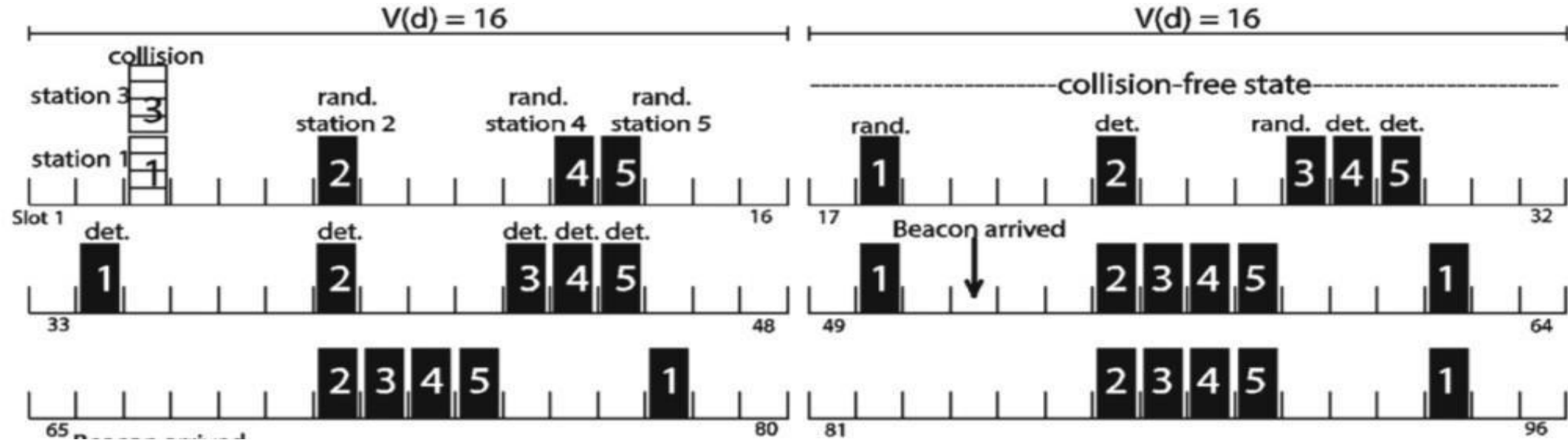
Beacon assisted Deterministic Backoff : Basic Concept



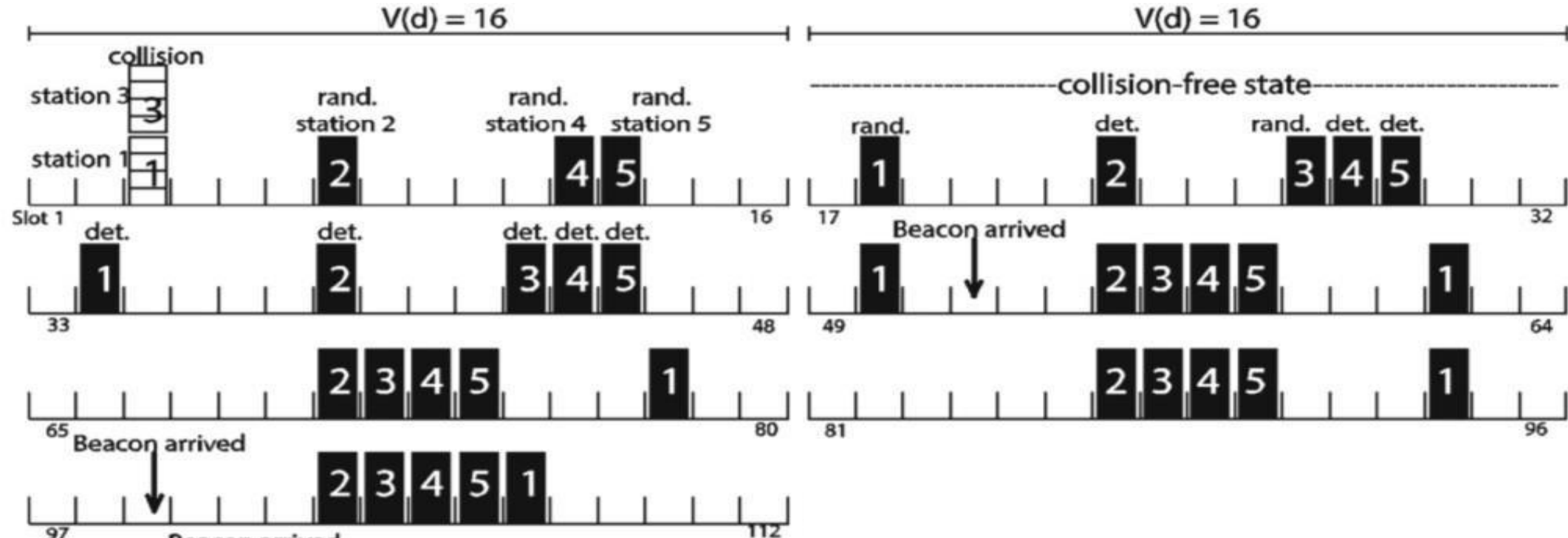
Beacon assisted Deterministic Backoff : Basic Concept



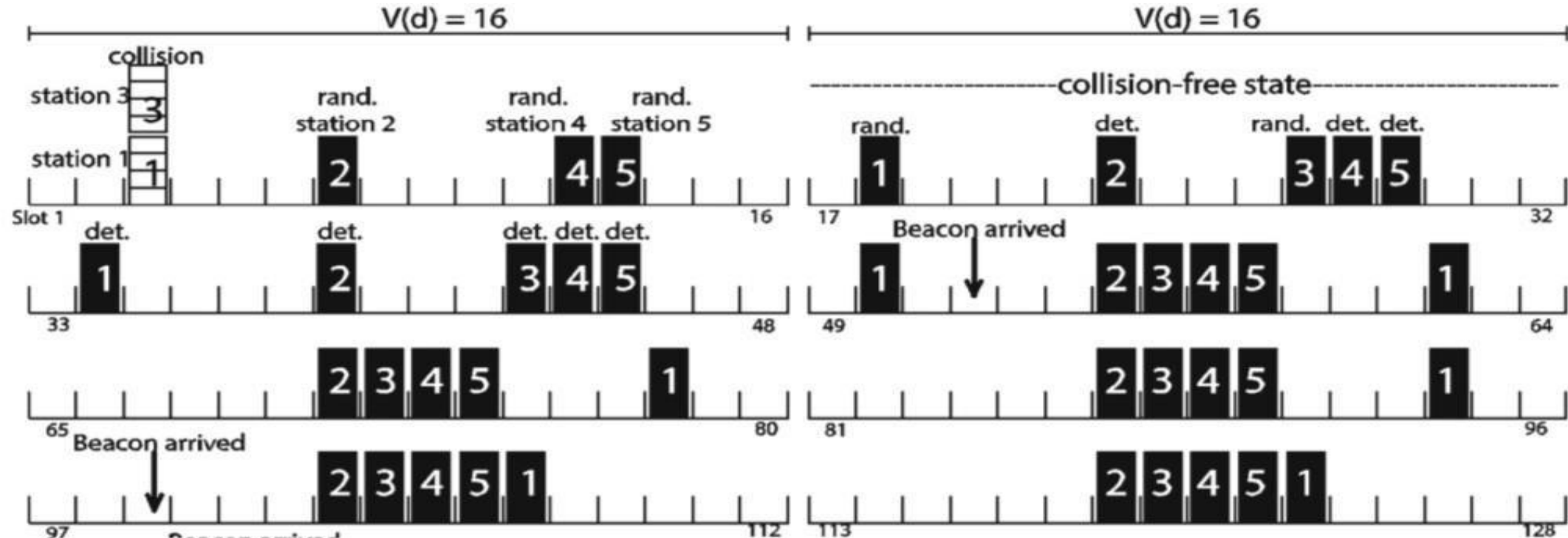
Beacon assisted Deterministic Backoff : Basic Concept



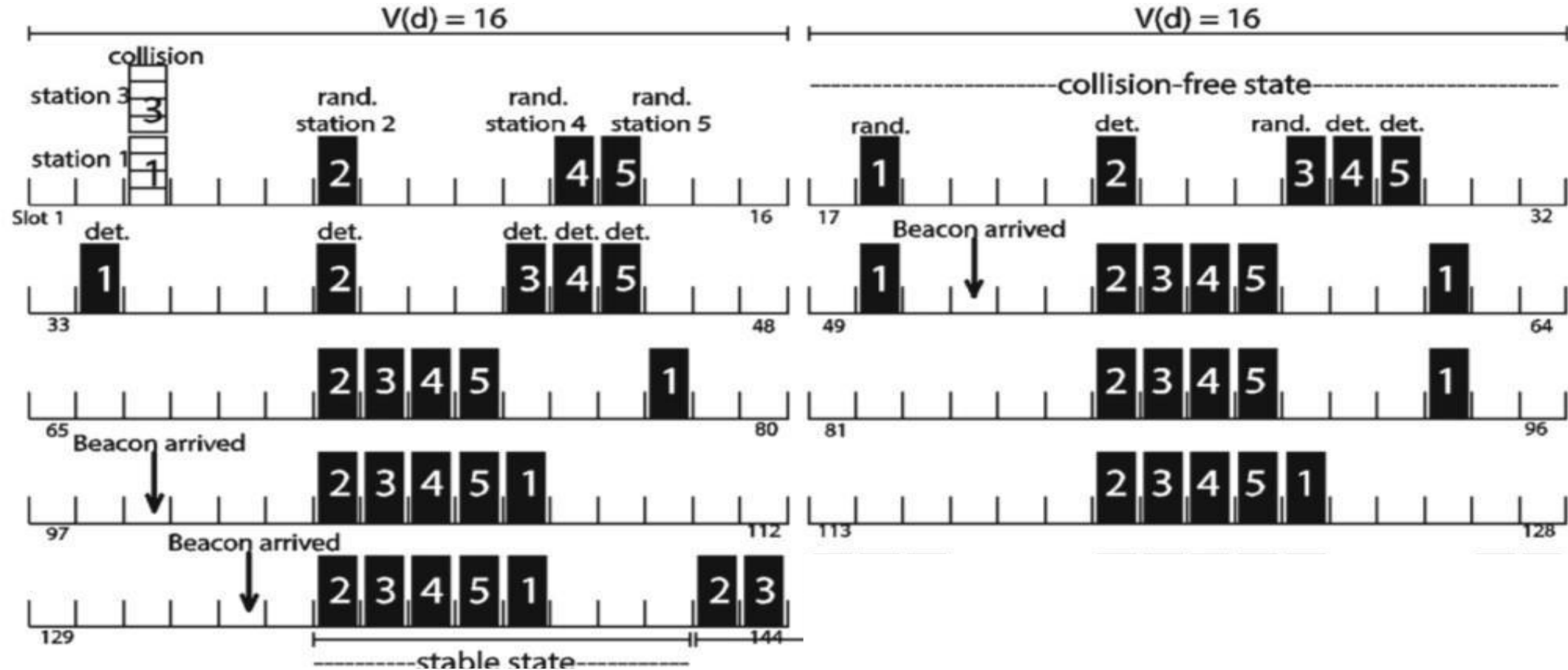
Beacon assisted Deterministic Backoff : Basic Concept



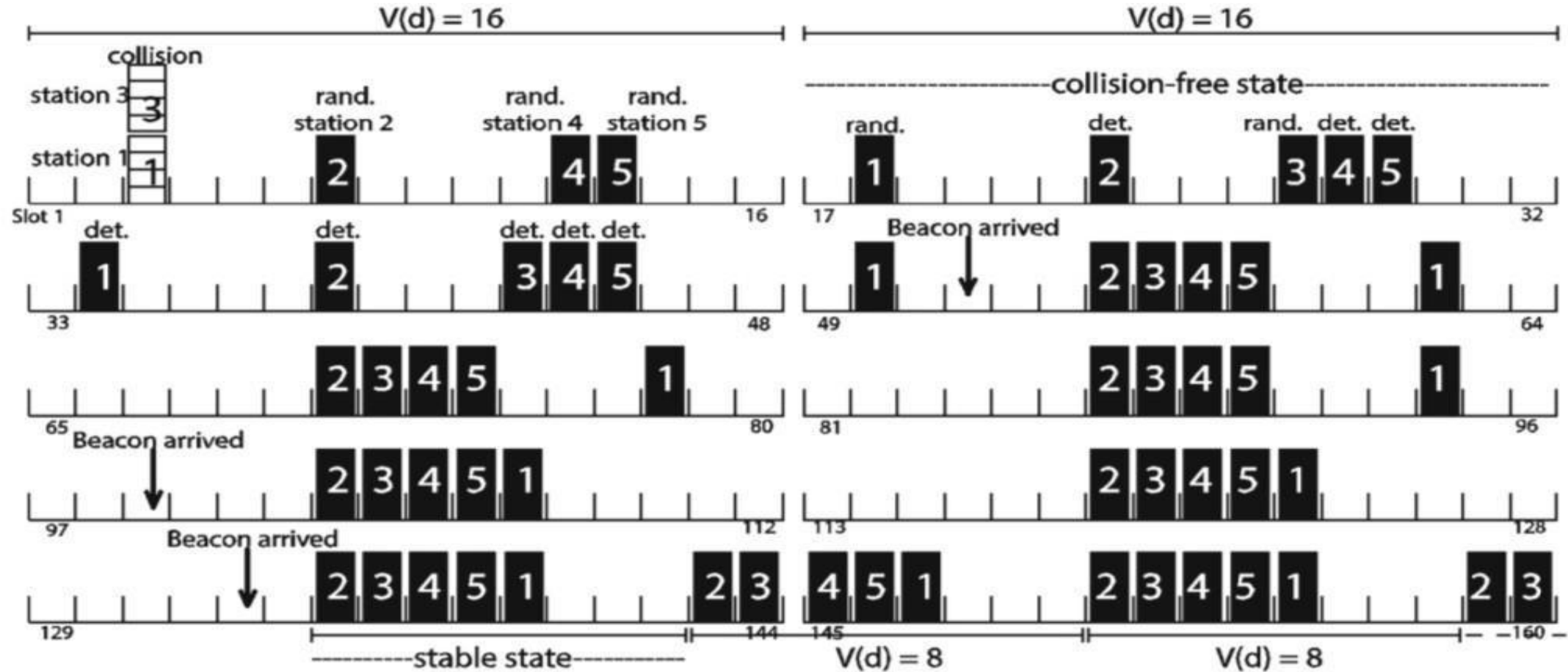
Beacon assisted Deterministic Backoff : Basic Concept



Beacon assisted Deterministic Backoff : Basic Concept



Beacon assisted Deterministic Backoff : Basic Concept



Beacon assisted Deterministic Backoff :: Strength

- **Its very Simple**
- **Doesn't require Special PHY (OFDM) Layer**
- **Doesn't require Additional Antenna at each Nodes**
- **Doesn't require all the nodes being tightly Time Synchronized**
- **Schedule preparation is very simple**
- **Less Channel wastage**
- **Scalable (Support a large number of contenders in a collision-free schedule)**
- **Converge to steady state is not very slow**
- **Handling the New node arrival and departure is trivial**
- **Coexistence and backwards compatibility with CSMA/CA**
- **Once the network converges to steady state, there will be no collisions and channel wastage will be minimum**

Beacon assisted Deterministic Backoff : Weaknesses

- Throughput efficiency may not reach optimal levels, but it is still significantly better than that of random-access mechanisms.
- Not highly efficient in highly dynamic networks, as newly arrived nodes can disrupt the network's steady state and cause instability again.
- **Required support of central coordinator/ AP for beacon transmission.**

Solution Approach -4

**A Distributed Approach for
Gradually Achieving Collision-Free Channel Access**

Paper Title: E-MAC: An evolutionary solution for collision avoidance in wireless ad hoc networks

Publisher: Journal of Network and Computer Applications (2016)

- Proposes a simple collision-avoidance MAC (E-MAC) for distributed wireless networks that can **iteratively achieve collision-free access.**
- In E-MAC, each transmitter will **adjust its next transmission time according to which part of its (just transmitted) packets suffering from the collision.**
- And the iteration of this **adjustment will quickly lead group of nodes converging to a collision-free network.**
- E-MAC **does not require any central coordination or global time synchronization.**
- It **is scalable to new entrants to the network and variable packet lengths**

Concept of collision-free Channel Access

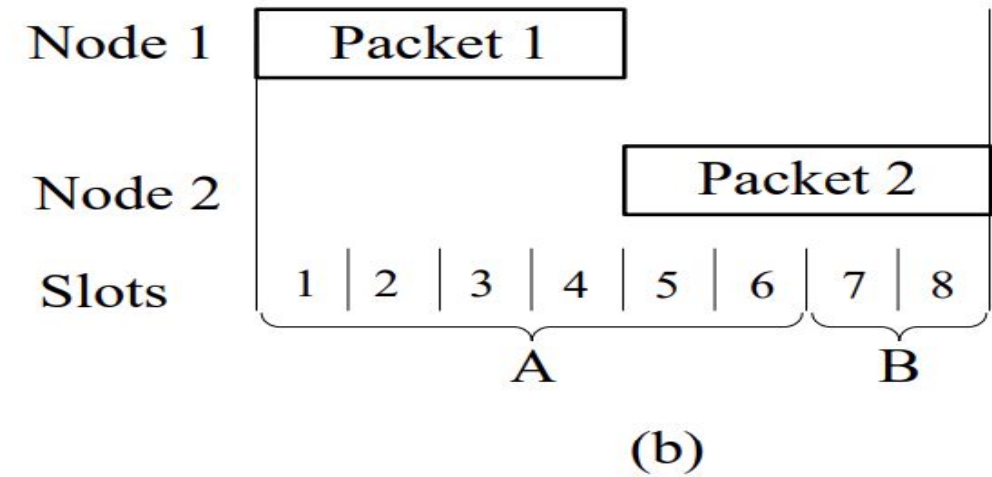
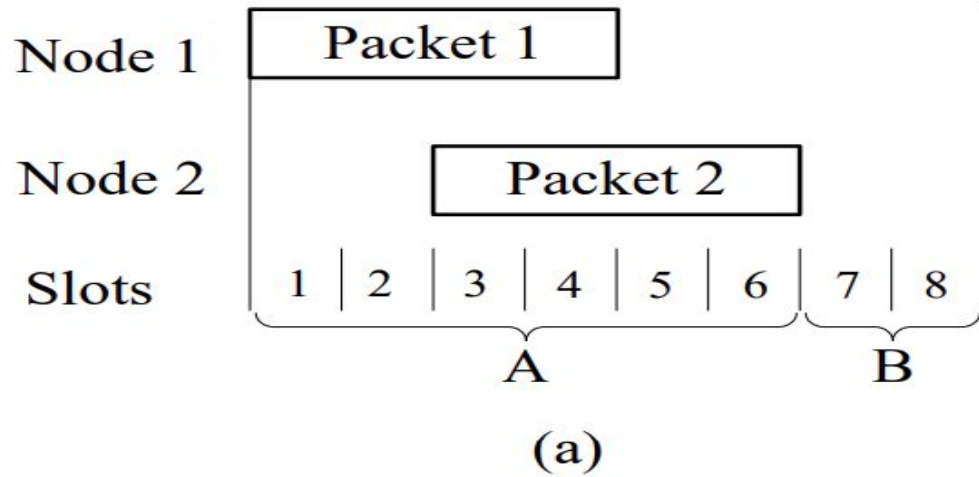


Fig.1. Illustrating the basic principle of packet collision

Packet Collision Scenarios

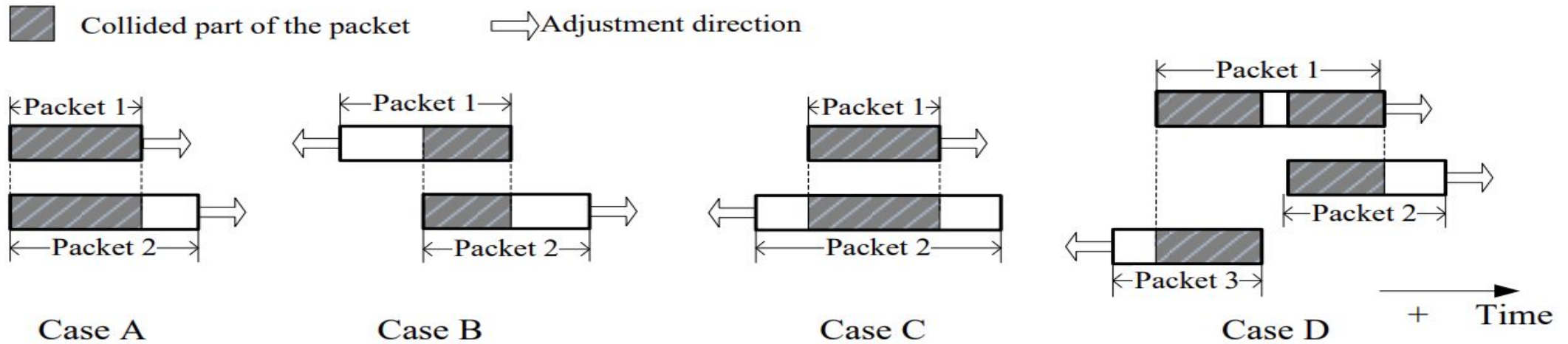
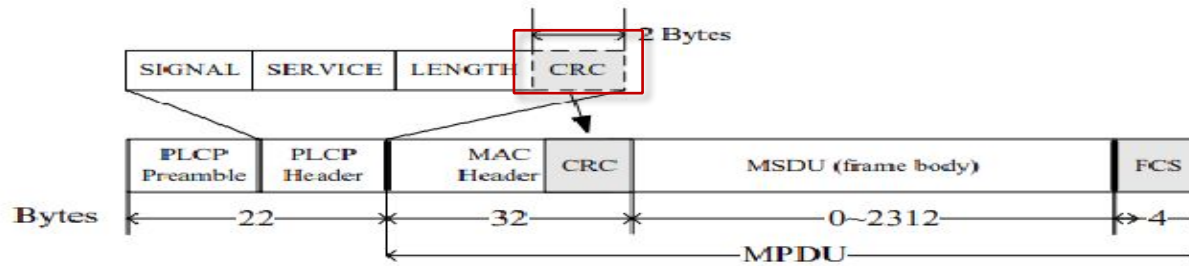


Fig. 2. Typical packet collision scenarios

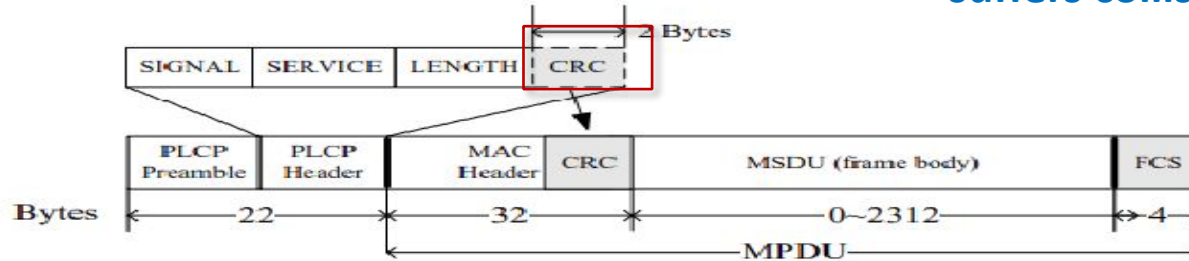
E-MAC: Revised Packet Format



(a) DATA

E-MAC: Revised Packet Format

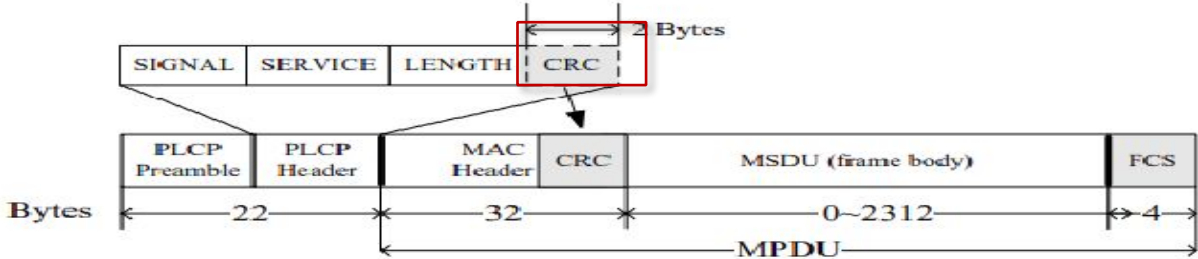
we can easily identify which part of the packet suffers collision for a collided packet



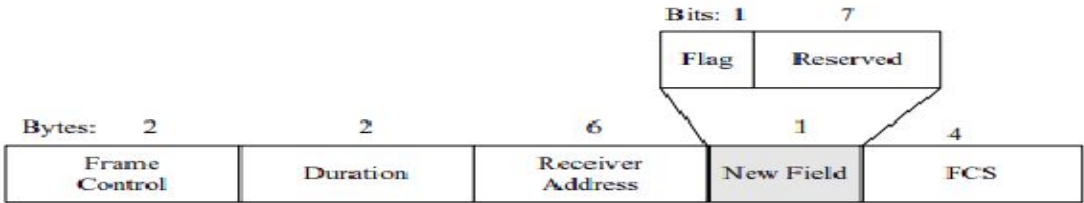
(a) DATA

E-MAC: Revised Packet Format

we can easily identify which part of the packet suffers collision for a collided packet



(a) DATA

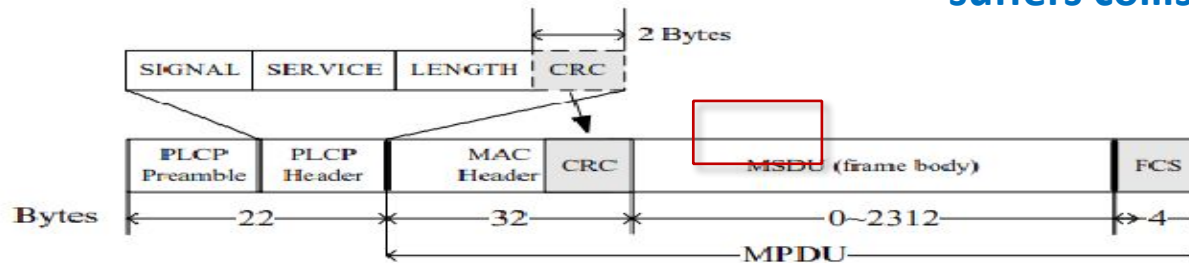


(b) NACK : Used to indicate “Collision but NOT on the Front part”

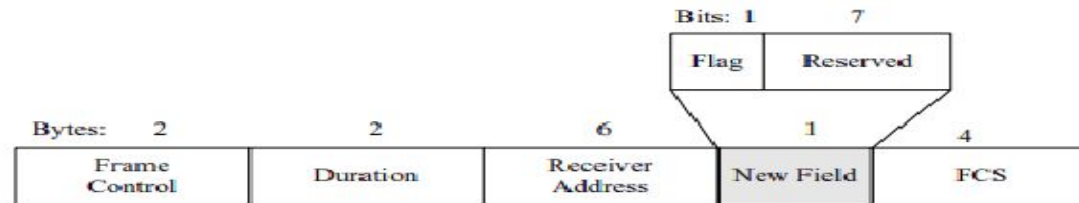
Fig. 5. Revision packet format

E-MAC: Revised Packet Format

we can easily identify which part of the packet suffers collision for a collided packet



(a) DATA

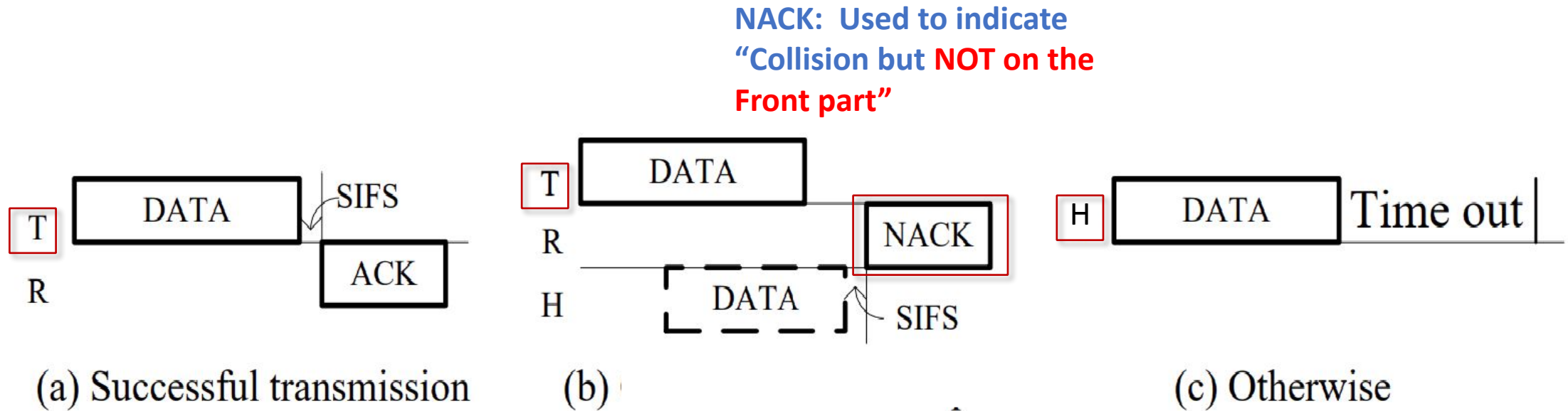


(b) NACK : Used to indicate "Collision but NOT on the Front part"

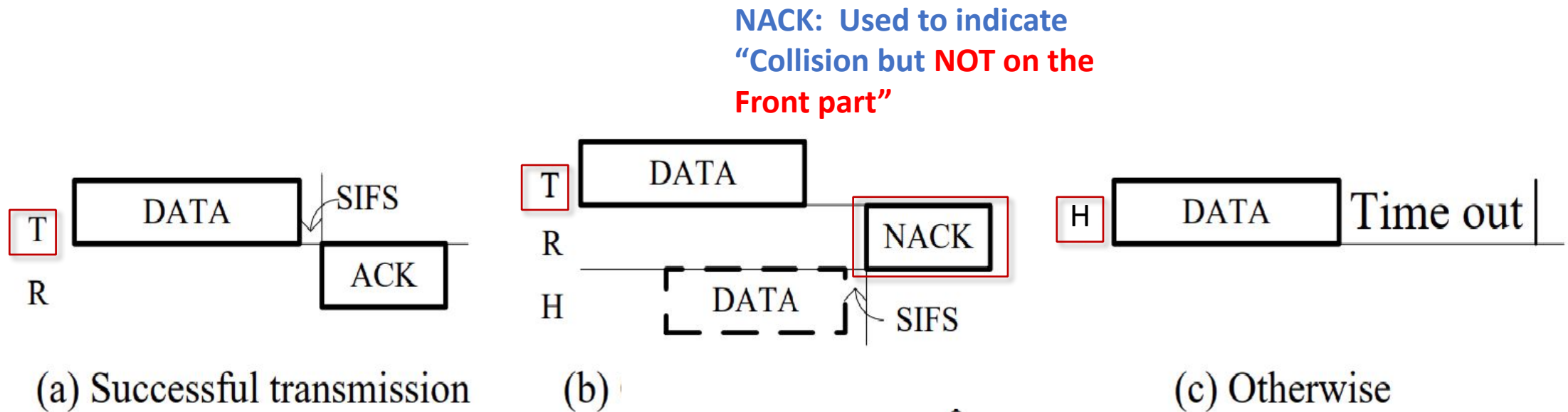
Fig. 5. Revision packet format

Supposition 1: For a collided packet, if the PLCP preamble, PLCP header, and MAC header are correctly decoded, this indicates that the collision did not occur in the front part of the packet.

E-MAC: 3 Types of Virtual Interpretations



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Supposition 1: For a collided packet, if the PLCP preamble, PLCP header, and MAC header are correctly decoded, this indicates that the collision did not occur in the front part of the packet.

E-MAC: Adaptive fine-grained Adjustment Step

3.3 The Proposed E-MAC Algorithm

Assuming that Node i is the next node to transmit. It will adjust its phase from ϕ_i to ϕ_i' as in (4), where $\alpha_i \in (0, 1/2]$ is a random variable.

$$\phi_i' = f(\Psi_i; \alpha_i) = \begin{cases} \phi_i, & \text{when Node } i \text{ experiences no collision} \\ \phi_i - \alpha_i l_i, & \text{when Node } i \text{ experiences a collision,} \\ & \text{but the front part of Node } i\text{'s packet is not collided} \\ \phi_i + \alpha_i l_i, & \text{otherwise} \end{cases} \quad (4)$$

Equation (4) indicates that if a collision occurs, the collided nodes will adjust their phases, namely the transmission time, according to which part of its transmitted packet suffers from the collision.

When no collision occurs, Node i will stick to its former phase. When a node experience a collision but

E-MAC: Adaptive fine-grained Adjustment Step

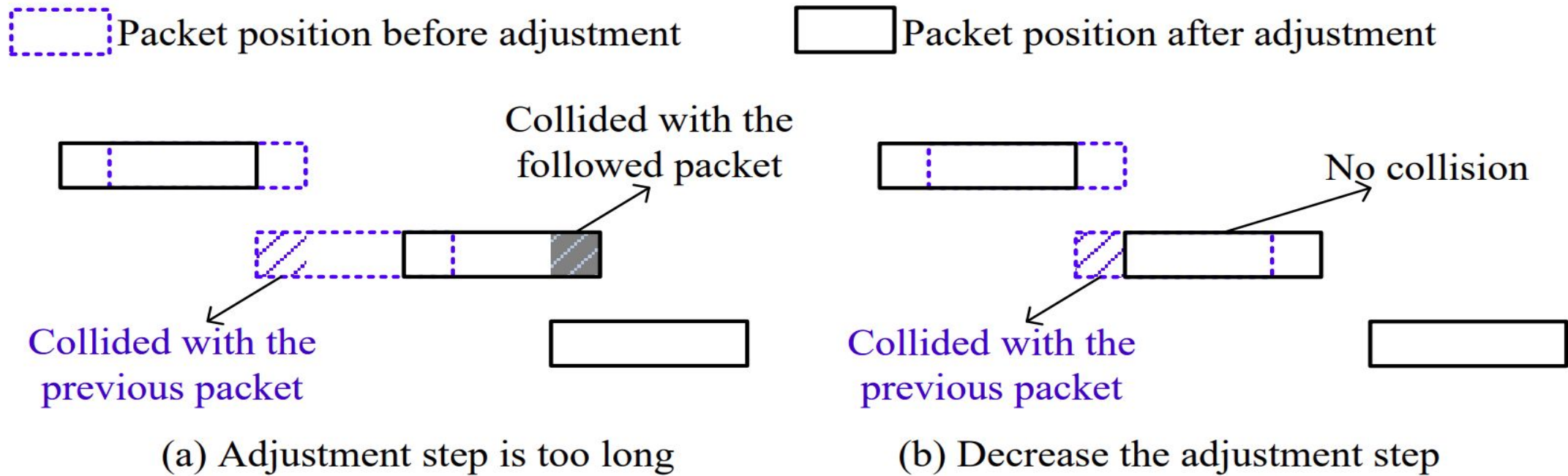
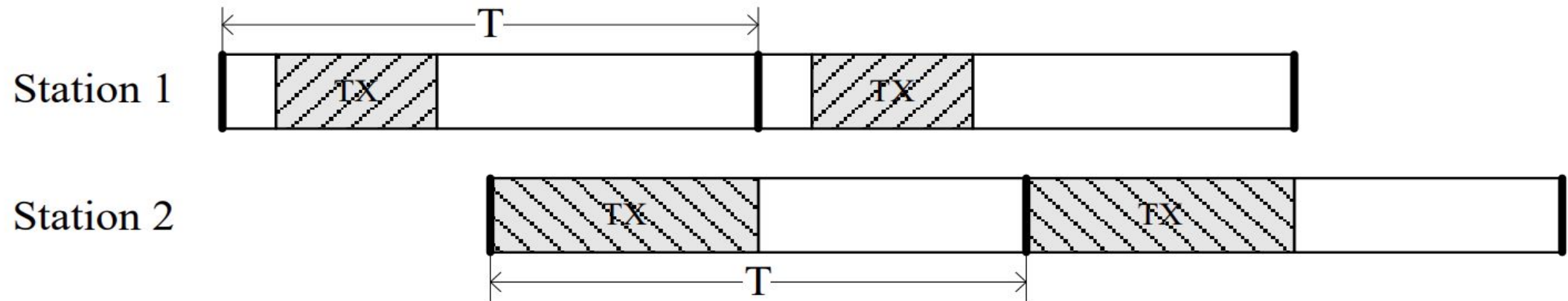
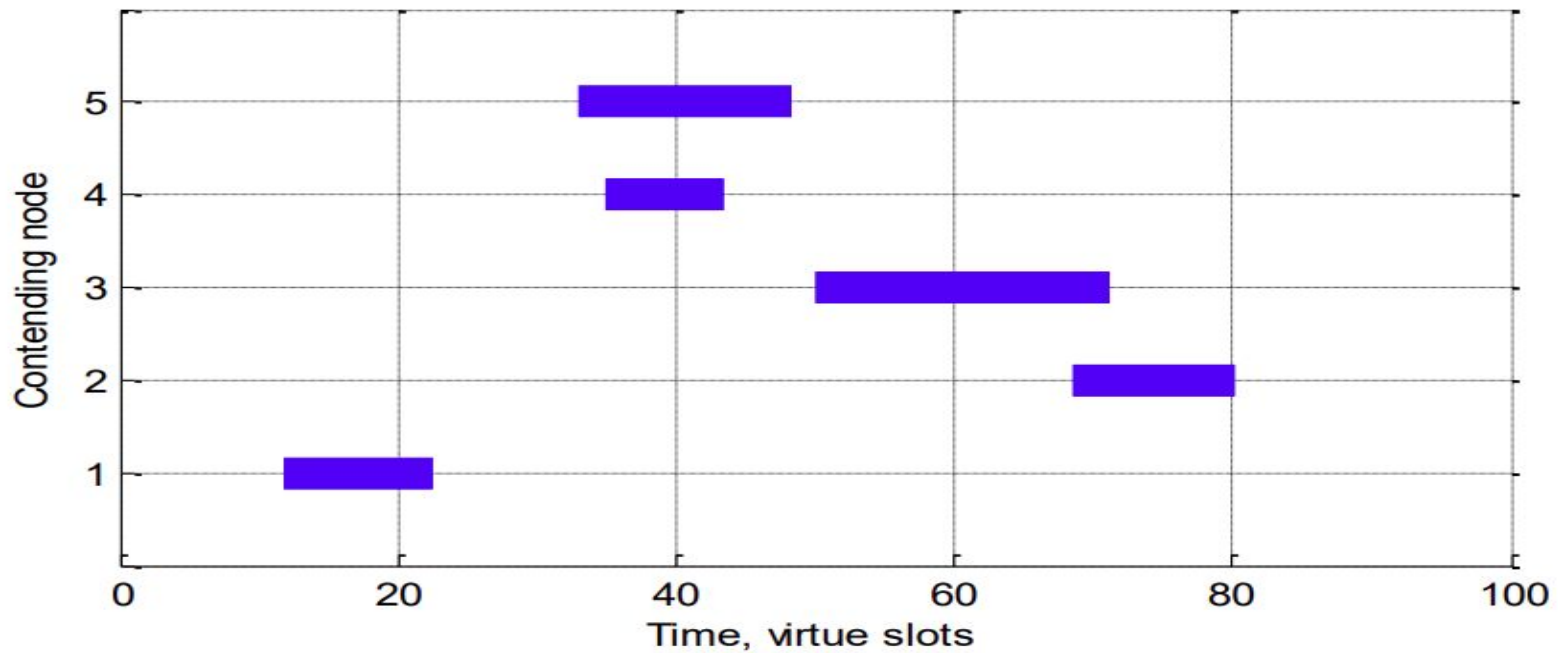


Fig. 7. Adaptive fine-grained adjustment step

E-MAC: **Ultimate** Collision-free Channel Access

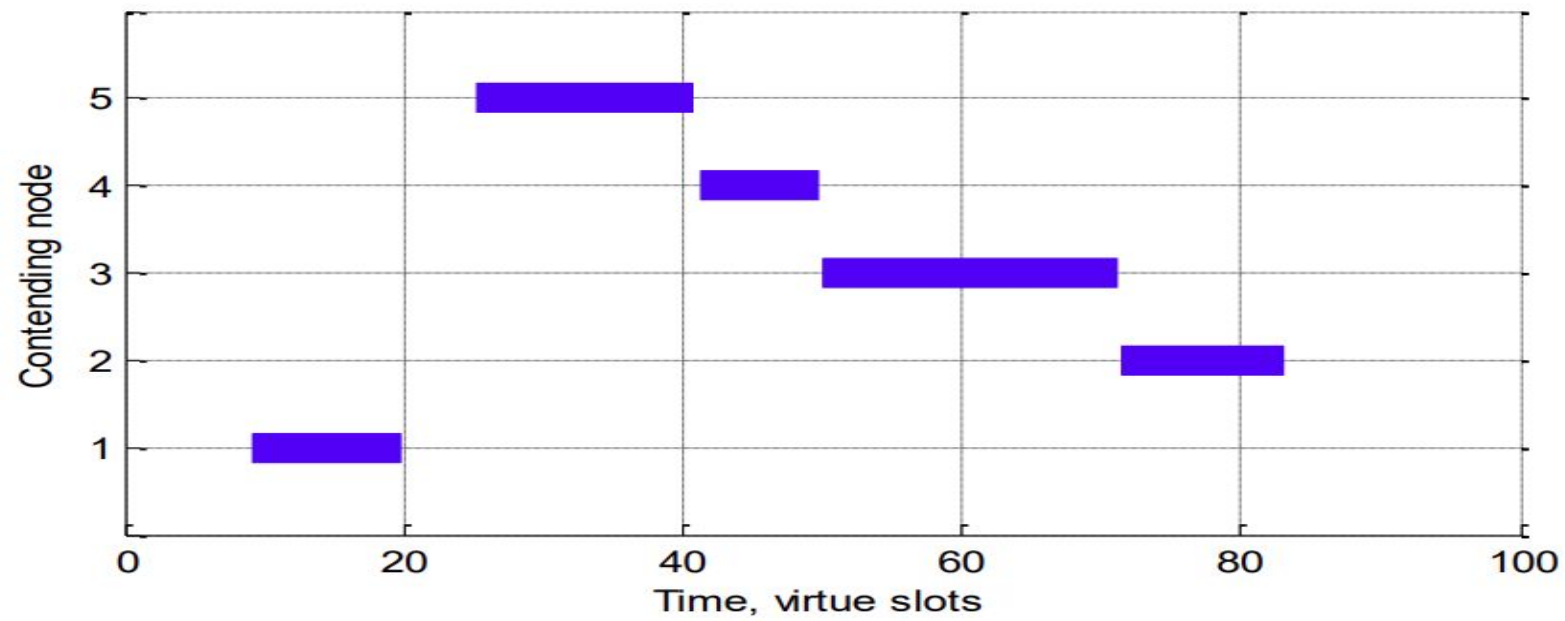


E-MAC: The initial transmissions with collision



(a) The initial transmissions with collision

E-MAC: The Achieved Collision-free State after m Iterations



(b) The achieved collision-free state after 5 iterations

Fig. 8. Illustrating the effectiveness of E-MAC

E-MAC:: Strength

- Doesn't require Special PHY (**OFDM**) Layer
- Doesn't require Additional Antenna at each Nodes
- Doesn't require any Central Coordination
- Doesn't require all the nodes being tightly Time Synchronized
- Doesn't require any dedicated Learning Phase
- Once the network converges to steady state, **there will be no collisions and channel wastage will be minimum**
- It is scalable to new entrants to the network and variable packet lengths.

E-MAC: Weaknesses

- **Learning Phase is still little bit complex**
- **Slow converges time (Converge to steady state is slow)**
- **Not efficient for Highly Dynamic Networks (newly arrived nodes may makes the steady state of the networks unsteady again)**

- Questions, comments?

- Thank you***